

Subspace embeddings with the rerandomized SRHT

Abstract

We study subspace embeddings obtained by two normalized real Walsh transforms, two independent sign diagonals, and uniform coordinate sampling without replacement. We prove that a sample size $k = O(r/\varepsilon^2)$, chosen independently of the target subspace, suffices to preserve all squared norms on each fixed r -dimensional subspace within $1 \pm \varepsilon$ with probability at least 0.99. The result holds for every ambient Walsh dimension and all ranks, and answers the without-replacement formulation of Problem 5.6 in the Simons workshop problem collection. The proof controls joint entry cumulants of the transformed projection through connected graph contractions and Walsh character identities. We then bound the expected trace of even powers of a product of centered projections. A two-projection decomposition converts this estimate into control of both spectral edges. The universal constant is not optimized.

1 Introduction and main result

An oblivious subspace embedding reduces dimension while approximately preserving all norms on a fixed subspace. Its distribution is chosen without knowing that subspace. Such maps provide a basic tool for randomized least-squares and low-rank approximation algorithms [10, 3]. For an r -dimensional subspace and fixed failure probability, Gaussian maps achieve squared-norm error ε with $O(r/\varepsilon^2)$ output coordinates [15, Section 1]. Fast structured maps seek the same geometric guarantee with less work when applying the map.

The subsampled randomized Hadamard transform (SRHT) applies random signs, a normalized Hadamard transform, and coordinate sampling. The fast Walsh–Hadamard transform permits multiplication in $O(n \log(2n))$ operations, compared with $O(nk)$ for a dense n -by- k map. Its one-round construction has a logarithmic obstruction: the coupon-collector construction of Halko, Martinsson, and Tropp for the Fourier transform [10, Remark 11.2], in Tropp’s Walsh formulation [16, Section 3.3], uses an r -dimensional subspace in ambient dimension r^2 and requires a sample of order $r \log r$ even for injectivity. The example concerns the traditional one-round law and motivates asking what a further randomization can accomplish.

The rerandomized SRHT in Problem 5.6 of the Simons workshop collection [1, Definitions 5.1 and 5.3, Problem 5.6] asks whether a second independent sign diagonal and Hadamard transform yield dimension proportional to r/ε^2 . We establish this bound for uniform sampling without replacement. We first specify the normalization and randomness used throughout the paper.

Let $n = 2^m$, $m \geq 0$, and identify the coordinate set with $K = (\mathbb{F}_2)^m$. The real normalized Walsh matrix is

$$H_{ab} = n^{-1/2}(-1)^{\langle a, b \rangle}, \quad H^T = H, \quad H^2 = I_n.$$

Let D_1, D_2 have mutually independent uniform Rademacher diagonal entries. Independently let J be a uniformly chosen k -element subset of K , and let S_J collect its coordinate vectors. Put

$$\Omega_{n,k} = \sqrt{\frac{n}{k}} D_1 H D_2 H S_J.$$

For a deterministic $n \times r$ frame V , $V^T V = I_r$, write

$$X = H D_2 H D_1 V, \quad P = X X^T, \quad \delta = \frac{r}{n}.$$

Thus $X^T X = I_r$, and the Gram matrix in question is

$$V^T \Omega_{n,k} \Omega_{n,k}^T V = \frac{n}{k} X^T S_J S_J^T X. \quad (1)$$

Here and below $\|\cdot\|_2$ denotes the Euclidean operator norm.

Theorem 1. *There is a universal constant $C \geq 1$ such that for every power of two n , every $1 \leq r \leq n$, and every $0 < \varepsilon < 1$, one can choose a deterministic integer*

$$r \leq k \leq \min\{n, \lceil Cr/\varepsilon^2 \rceil\}$$

independently of V , for which

$$\sup_{V^T V = I_r} \mathbb{P}\left\{\|V^T \Omega_{n,k} \Omega_{n,k}^T V - I_r\|_2 > \varepsilon\right\} \leq 0.01. \quad (2)$$

For a fixed frame V , the event complementary to that in (2) says precisely that

$$(1 - \varepsilon)\|z\|_2^2 \leq \|\Omega_{n,k}^T V z\|_2^2 \leq (1 + \varepsilon)\|z\|_2^2 \quad \text{for every } z \in \mathbb{R}^r.$$

The supremum in (2) is outside the probability: the frame is fixed before both sign diagonals and the sample, while the chosen width depends only on n, r, ε . The theorem selects a width with the stated guarantee; it does not require monotonicity of the normalized failure probability at every larger width. Full sampling gives the Gram matrix I_r exactly. The assumption $n = \Omega(\log r)$ printed in Problem 5.6 imposes no additional restriction here because $\log r \leq r \leq n$.

Proof overview

We first replace the fixed-size sample by a Bernoulli coordinate projection E , independent of $P = X X^T$, whose diagonal entries have mean $0 < \theta \leq 1/2$. The independent selectors make it possible to expand the trace by grouping equal coordinate indices. The normalized Gram error is $\theta^{-1} X^T (E - \theta I_n) X$. A direct trace expansion therefore involves entries of P . Since $\mathbb{E} P = \delta I_n$, with $\delta = r/n$, we also center P and set

$$R = P - \delta I_n, \quad W = E - \theta I_n, \quad A = RW.$$

This second centering removes every first-order entry cumulant, including the diagonal ones. Consequently, the cumulant expansion contains only blocks of at least two entries, a restriction needed for the later counts.

Centering both factors leaves a product A that is generally nonsymmetric, and its trace at an even order $2p$ can be negative. Lemma 2 deals with this difficulty deterministically. The two-projection decomposition expresses the compressed Gram eigenvalues through scalar and two-dimensional invariant blocks. In each two-dimensional block of A , the two roots have a fixed product, while their sum changes with the Gram eigenvalue. This gives the lower bound $\text{tr } A^{2p} \geq -2r[\delta(1 - \delta)\theta(1 - \theta)]^p$. After adding this deterministic compensation, a deviation at either spectral edge forces a large trace contribution, under the hypotheses of that lemma. Thus it suffices to bound the signed expectation $\mathbb{E} \text{tr } A^{2p}$; no estimate of $\mathbb{E} |\text{tr } A^{2p}|$ is required.

The random part of the proof must control both the size and the number of terms in this trace expansion. Section 3 first bounds the joint entry cumulants. The product formula in Lemma 6 expands the two independent sign layers over connected partitions. Each resulting graph sum contains a rank- r projection edge; factoring it through $\mathbb{R}^r$ allows Lemma 5 to bound the sum by r . Lemma 7 combines this estimate with Walsh character symmetry, which makes a cumulant vanish unless its endpoint labels satisfy a binary equation. Section 4 then counts the terms that can remain. Lemma 8 bounds the equality patterns of the sampling indices, Lemma 9 counts labels satisfying the binary equations, and Proposition 10 bounds the contribution of one equality pattern. Each loss of an equality block removes a free row label together with its selector factor, giving $(n\theta)^{-1}$. Losing a cumulant block gives r^{-1} , and an independent binary equation gives n^{-1} . Keeping these factors in the count offsets its growth with the moment order and yields Proposition 3.

Combining that trace estimate with the deterministic transfer gives, under the hypotheses of Lemma 4,

$$\mathbb{P}\{\|\theta^{-1}X^T EX - I_r\|_2 > \eta\} \leq (K_0^p + 2r) \left(\frac{4r}{\kappa\eta^2} \right)^p, \quad \kappa = n\theta,$$

where $0 < \eta < 1$, κ is the expected sample size, and K_0 is the universal constant in Proposition 3. When κ is a sufficiently large constant multiple of r/η^2 , both terms on the right decay geometrically in p , apart from the factor r . A moment order proportional to $\log r$ absorbs this factor without introducing an extra logarithm into the sampling width. The rank condition in Lemma 4 restricts this argument to sufficiently large r .

Section 5 completes the sampling and parameter choices. Lemma 11 couples a uniform fixed-size sample between two Bernoulli samples; set inclusion compares their Gram matrices, and binomial tails control the sample counts. For the remaining ranks, Lemma 12 averages over both rerandomization and sampling to obtain a second-moment bound with width of order r^2/ε^2 . These ranks lie below a fixed universal threshold R_0 , so $r^2 \leq R_0 r$ absorbs this width into the theorem's universal constant. Full sampling gives an exact Gram matrix whenever the ambient dimension is too small for the large-rank coupling. All choices of width are independent of V . Figure 1 records the dependencies among the numbered results.

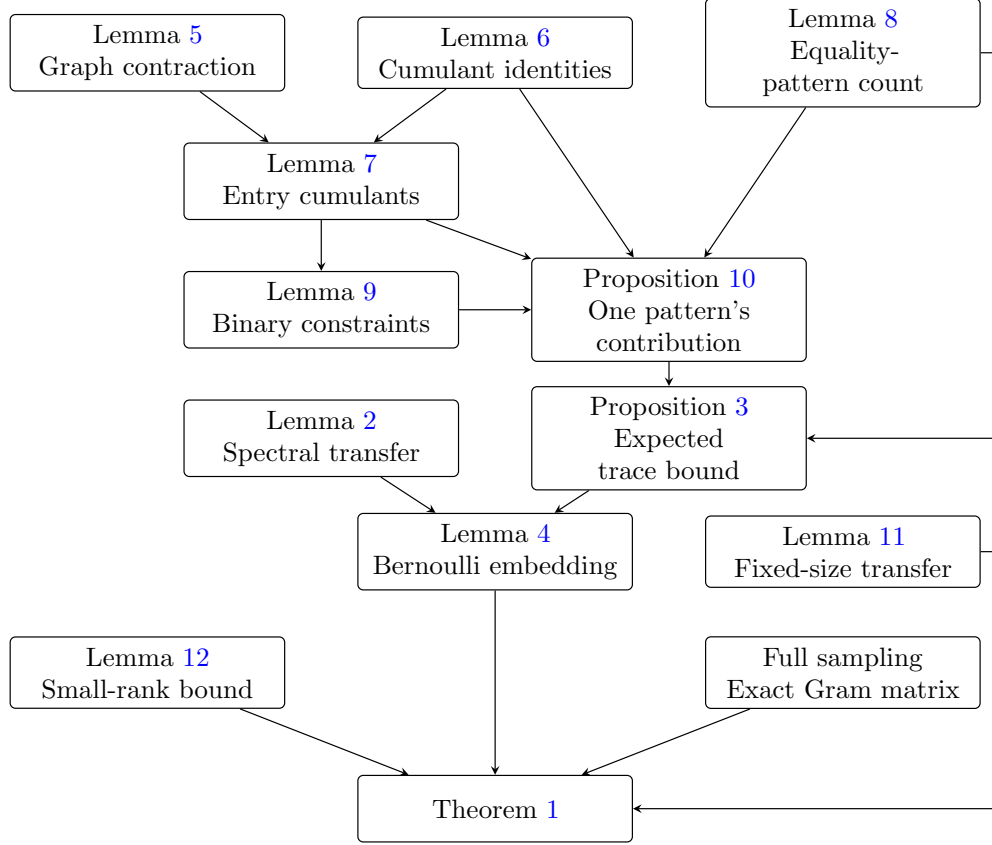


Figure 1: Proof dependencies. An arrow points from a result to a proof that uses it. Theorem 1 combines the Bernoulli estimate and fixed-size transfer for large ranks, the second-moment bound for small ranks, and full sampling when needed. Definitions and routine calculations are omitted.

Relation to prior work

The dimension benchmark concerns all oblivious linear maps, including dense ones. For $0 < \varepsilon < 1/3$ and fixed failure probability 0.01, Nelson and Nguyen [15, Theorem 6] prove the lower bound

$$k = \Omega(\min\{n, r/\varepsilon^2\}).$$

Their theorem uses relative norm error; squared-norm error ε implies that norm guarantee with the same ε . The Gaussian upper bound, with a constant rescaling of its accuracy parameter, attains the r/ε^2 order. The identity map covers ambient-dimension saturation [15, Section 1]. This comparison explains both the dimension scale in Problem 5.6 and the necessity of retaining the cap by n .

Fast structured maps are particularly useful when applying a dense sketch would dominate the matrix computation. Halko, Martinsson, and Tropp [10, Sections 4.6 and 11] develop structured random test matrices for approximate matrix decompositions. Boutsidis and Gittens [3, Theorems 2.1 and 3.1, Lemma 4.11] analyze SRHT low-rank approximation, least-squares regression and matrix multiplication. For subspace embeddings, Tropp’s analysis [16, Theorem 3.1] combines row-norm control with matrix Chernoff bounds.

Multiple Hadamard/sign rounds have also been used for other objectives. Andoni, Indyk, Laarhoven, Razenshteyn, and Schmidt [2, Section 3.1] use a three-round transform as a fast rotation in cross-polytope locality-sensitive hashing. Choromanski et al. [6, Lemma 1 and Theorems 5.1–5.3] give distributional and hashing guarantees for three-round structures within the TripleSpin

framework. These are the multi-round constructions discussed after Problem 5.6 in [1]; their conclusions concern different observables from the uniformly sampled subspace Gram matrix. For the same unsampled two-round rotation used here, Zilca and Mendelson [18, Equation (1) and Theorem 4.1] prove an $O(n^{-1/5})$ coordinate Kolmogorov bound relative to a Haar-rotated vector, uniformly over fixed unit inputs and coordinate indices. Their nonvanishing worst-case full-vector Wasserstein discrepancy [18, Theorems 5.1–5.2] addresses the full output distribution. Theorem 1 instead controls the two spectral edges after coordinate sampling.

Other random-map laws achieve linear dependence on the subspace dimension. Chenakkod, Dereziński, Dong, and Rudelson [5, Theorem 4.5] combine a Hadamard preprocessing step with an independent sparse factor. With singular-value error $\eta = \varepsilon/3$ and failure probability 0.01, their theorem gives dimension $O(r/\varepsilon^2)$ in the regime $r > \log(200n)$, with its prescribed sparsity parameters. The sparse factor takes the place of the second Walsh transform and uniform coordinate subset in the present construction.

A related set of proof tools appears in the work of Chenakkod, Dereziński, and Dong [4, Theorems 3 and 7]. For $r > 10$, they study OSNAP embeddings and obtain dimension $O(r/\varepsilon^2)$ with near-optimal column sparsity when the inverse distortion and inverse failure probability are at most polynomial in r . Their analysis combines decoupling, cumulant expansions along Gaussian interpolation, and trace inequalities [4, Section 4; full version, Sections 7.2–7.5]. In our fixed Walsh model, connected graph sums and character identities control the joint entry cumulants directly. The subsequent trace is that of a generally nonsymmetric product of centered projections; the two-projection calculation provides the compensation required to recover both spectral edges.

Centered alternating traces of projections also appear in Kunisky’s analysis of the MANOVA law, a limiting eigenvalue distribution for products of random projections [11, Theorems 1.5 and 1.12]. His moment recursions give sufficient conditions for convergence in probability of the largest eigenvalue. The common starting point is the centered trace; here the spectral transfer is a finite-dimensional inequality with explicit compensation, and the moment estimate must hold for the prescribed two-sign Walsh model.

Vázquez-Becerra [17, Section 4] combines cumulant expansions, graph sums, and the bounds of Mingo and Speicher to study fluctuations of traces, including ensembles built from discrete Fourier matrices. These tools are closely related to those in Section 3. The distinction needed here is control of fixed matrix entries, with explicit dependence on cumulant order and on the rank of an arbitrary fixed input projection. The remarks after Lemmas 2 and 7 give the corresponding method comparisons.

The remainder is organized as follows. Section 2 gives the deterministic spectral reduction and the Bernoulli embedding consequence of the trace estimate. Section 3 proves the joint cumulant bounds, and Section 4 supplies the complete trace calculation. Section 5 treats fixed-size sampling and all remaining ranks and dimensions to prove Theorem 1. Throughout, unadorned logarithms are natural, $\log_2$ is the binary logarithm, and $\|\cdot\|_F$ denotes the Frobenius norm.

2 From spectral deviations to a centered trace estimate

We begin with Bernoulli sampling. Let E be a diagonal projection whose entries are independent Bernoulli variables of mean θ , independent of the two sign diagonals. The matrix $\theta^{-1}X^T E X$ is the corresponding normalized Gram matrix. Since $X^T X = I_r$, its deviation is

$$\theta^{-1}X^T E X - I_r = \theta^{-1}X^T (E - \theta I_n) X. \quad (3)$$

Writing $W = E - \theta I_n$, cyclicity of the trace gives, for every positive integer j ,

$$\operatorname{tr}(X^T W X)^j = \operatorname{tr}(P W)^j, \quad P = X X^T.$$

Thus a direct moment expansion introduces entries of the uncentered projection P . We will prove in Lemma 7 that $\mathbb{E}P = \delta I_n$. Subtracting this mean removes every first-order entry cumulant, including those of diagonal entries. This is the reason for also centering P : the entry-cumulant expansion will then contain only blocks of size at least two. The resulting product need not be symmetric, so a signed trace estimate requires a separate passage back to the Gram eigenvalues. Write $R = P - \delta I_n$, $W = E - \theta I_n$, and $A = RW$. This section shows why controlling a signed even trace of A suffices to bound deviations of that Gram matrix.

The first step is deterministic. For two orthogonal projections, the compressed eigenvalues describe two-dimensional invariant blocks. Centering the projections changes each block to a matrix with a quadratic characteristic polynomial. Its real and nonreal roots contribute differently to an even trace; the following lemma keeps the possible negative contribution explicit.

Lemma 2. *Let $P = X X^T$ and E be any real orthogonal projections, $X^T X = I_r$. Let $0 < \delta, \theta \leq 1/2$ and set*

$$A = (P - \delta I)(E - \theta I), \quad a = \sqrt{\delta(1 - \delta)\theta(1 - \theta)}.$$

For every integer $p \geq 1$, $\operatorname{tr} A^{2p} \geq -2ra^{2p}$. If $0 < \eta < 1$ and $\delta \leq \theta\eta^2/64$, then pointwise

$$1_{\{\|\theta^{-1}X^T E X - I_r\|_2 > \eta\}}(\theta\eta/2)^{2p} \leq \operatorname{tr} A^{2p} + 2ra^{2p}. \quad (4)$$

Proof. Fix an integer $p \geq 1$. We use a common invariant-block decomposition of P and E to control cancellation in $\operatorname{tr} A^{2p}$. The fixed eigenvalue product on each two-dimensional block bounds its possible negative contribution. Under the additional small- δ hypothesis, a spectral deviation forces one block to contribute more than $(\theta\eta/2)^{2p}$. The blockwise lower bounds control the remaining contributions and yield the stated inequality.

Compressed spectrum. The compressed operator is PEP restricted to range P . Since $X^T X = I_r$ and $P = X X^T$, the map $v \mapsto Xv$ is an isometry onto that space, and

$$PEPX = X(X^T E X).$$

Thus its eigenvalues $\lambda_1, \dots, \lambda_r$ are precisely those of $X^T E X$. They lie in $[0, 1]$, because $0 \leq u^T E u \leq \|u\|^2$. In particular,

$$\|\theta^{-1}X^T E X - I_r\|_2 = \theta^{-1} \max_{1 \leq i \leq r} |\lambda_i - \theta|.$$

Invariant blocks. We use the standard two-projection decomposition [9, Theorem 2], constructing it directly in finite dimensions to include the endpoint eigenspaces. Diagonalize PEP on range P . For a unit eigenvector u with eigenvalue $0 < \lambda < 1$, put

$$w = \frac{Eu - \lambda u}{\sqrt{\lambda(1 - \lambda)}}.$$

Here $\lambda u = PEu$, so w is the normalized component of Eu perpendicular to range P . Indeed, $\|Eu\|^2 = \langle u, Eu \rangle = \lambda$, and hence

$$P(Eu - \lambda u) = 0, \quad \|Eu - \lambda u\|^2 = \lambda - 2\lambda^2 + \lambda^2 = \lambda(1 - \lambda).$$

It follows that $Pw = 0$, $\|w\|_2 = 1$, and u, w are orthonormal. The definition of w and $E^2 = E$ give

$$Eu = \lambda u + \sqrt{\lambda(1-\lambda)} w, \quad Ew = \frac{(1-\lambda)Eu}{\sqrt{\lambda(1-\lambda)}} = \sqrt{\lambda(1-\lambda)} u + (1-\lambda)w.$$

Thus their span is invariant under both projections, whose matrices in this basis are

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad E = \begin{pmatrix} \lambda & \sqrt{\lambda(1-\lambda)} \\ \sqrt{\lambda(1-\lambda)} & 1-\lambda \end{pmatrix}. \quad (5)$$

Distinct eigenbasis vectors u, u' with eigenvalues $\lambda, \lambda' \in (0, 1)$ give orthogonal pairs, even when $\lambda = \lambda'$. To see this, use

$$\langle u, Eu' \rangle = \langle u, PEPu' \rangle = \lambda' \langle u, u' \rangle = 0, \quad \langle Eu, Eu' \rangle = \langle u, Eu' \rangle = 0.$$

Together with the definitions of w, w' , these identities give $\langle u, w' \rangle = \langle w, u' \rangle = \langle w, w' \rangle = 0$.

For the endpoint eigenvalues $\lambda = 0, 1$, no second vector w is needed. In either case $Pu = u$. If $\lambda = 0$, then $\|Eu\|^2 = \lambda = 0$, so $Eu = 0$. If $\lambda = 1$, then $\|(I - E)u\|^2 = 1 - \lambda = 0$, so $Eu = u$. The corresponding eigenspaces are therefore $\text{range } P \cap \ker E$ and $\text{range } P \cap \text{range } E$, respectively. Each endpoint eigenvector in our orthonormal eigenbasis spans a one-dimensional subspace invariant under both P and E .

We have now placed every vector of the eigenbasis of $\text{range } P$ in a one- or two-dimensional invariant block. It remains to handle the orthogonal complement of all these blocks. Because the blocks together contain $\text{range } P$, this complement lies in $\ker P$, so P is zero there. To check that E preserves it, take a vector v in the complement and a vector z in any constructed block. Then

$$\langle Ev, z \rangle = \langle v, Ez \rangle = 0,$$

where the first equality uses symmetry of E , and the last follows because Ez stays in the same block. Thus the restriction of E to the complement is again an orthogonal projection. An orthonormal eigenbasis of this restriction splits the complement into one-dimensional blocks on which $P = 0$ and $E = 0$ or $E = 1$. This completes the decomposition of the entire ambient space.

Finally, a two-dimensional block is created only for an eigenbasis vector whose eigenvalue lies in $(0, 1)$. There are r vectors in the eigenbasis of $\text{range } P$, so the number of two-dimensional blocks is at most r .

Block polynomial. On (5), the restriction of A is

$$\begin{pmatrix} (1-\delta)(\lambda-\theta) & (1-\delta)\sqrt{\lambda(1-\lambda)} \\ -\delta\sqrt{\lambda(1-\lambda)} & -\delta(1-\lambda-\theta) \end{pmatrix}.$$

Its trace is $(1-\delta)(\lambda-\theta) - \delta(1-\lambda-\theta) = \lambda - \delta - \theta + 2\delta\theta$. Its determinant is a^2 : the two centered factors have determinants $-\delta(1-\delta)$ and $-\theta(1-\theta)$. Consequently its characteristic polynomial is

$$z^2 - tz + a^2, \quad t = \lambda - \delta - \theta + 2\delta\theta. \quad (6)$$

Write its roots as z_1, z_2 . The product and sum of the roots satisfy

$$z_1 z_2 = a^2, \quad z_1 + z_2 = t = (\lambda - \theta) - \delta(1 - 2\theta).$$

Thus the product is independent of λ , while the sum records the deviation $\lambda - \theta$ up to the shift $-\delta(1 - 2\theta)$.

Trace lower bound. To prove $\text{tr } A^{2p} \geq -2ra^{2p}$, consider a two-dimensional block. Its contribution to $\text{tr } A^{2p}$ equals $z_1^{2p} + z_2^{2p}$, including when the block has a defective repeated root. Indeed, Cayley–Hamilton gives the recurrence $S_j = tS_{j-1} - a^2S_{j-2}$, $j \geq 2$, for both the power traces and the root-power sums, with the same initial values $S_0 = 2$, $S_1 = t$.

If the roots are nonreal, they are conjugates, and $z_1z_2 = |z_1|^2 = a^2$ gives

$$z_1^{2p} + z_2^{2p} = 2\text{Re}(z_1^{2p}) \geq -2a^{2p}.$$

If the roots are real, their even-power sum is nonnegative. The remaining one-dimensional eigenvalues are among

$$-(1-\delta)\theta, \quad (1-\delta)(1-\theta), \quad -\delta(1-\theta), \quad \delta\theta,$$

corresponding to the four possibilities $(P, E) = (1, 0), (1, 1), (0, 1), (0, 0)$ on a common eigenvector. Their even powers are nonnegative. Summing over at most r two-dimensional blocks proves $\text{tr } A^{2p} \geq -2ra^{2p}$.

Spectral deviation. For the second conclusion, assume $0 < \eta < 1$ and $\delta \leq \theta\eta^2/64$. Suppose the bad event in (4) occurs. Since the deviation norm equals $\theta^{-1} \max_i |\lambda_i - \theta|$, there is a compressed eigenvalue with $|\lambda - \theta| > \theta\eta$. Set $b = \theta\eta$. We will find a block contributing more than $(b/2)^{2p}$.

For $0 < \lambda < 1$, the corresponding block has polynomial (6). The hypothesis and $0 < \eta < 1$ give $\delta \leq b\eta/64 < b/64$. Since $0 \leq 1 - 2\theta \leq 1$,

$$\begin{aligned} |t| &= |(\lambda - \theta) - \delta(1 - 2\theta)| \geq |\lambda - \theta| - \delta(1 - 2\theta) > b - \delta > 63b/64, \\ a^2 &\leq \delta\theta \leq \theta^2\eta^2/64 = b^2/64. \end{aligned}$$

In particular,

$$|t| > 63b/64, \quad a \leq \sqrt{\delta\theta} \leq b/8.$$

It follows that $|t| > 2a$, so the discriminant $t^2 - 4a^2$ is positive and both roots are real. They have the same sign because $z_1z_2 = a^2 > 0$. Also,

$$\min\{|z_1|, |z_2|\} \leq \sqrt{|z_1z_2|} = a, \quad |z_1| + |z_2| = |t|.$$

Thus the larger absolute value is at least $|t| - a$. Since $|t| - a > 55b/64 > b/2$ and both even powers are nonnegative, this block contributes more than $(b/2)^{2p}$.

The endpoint cases give the same conclusion in one dimension. For $\lambda = 0$, the eigenvalue of A is $-(1-\delta)\theta$, with magnitude $(1-\delta)\theta \geq \theta/2 > b/2$. For $\lambda = 1$, it is $(1-\delta)(1-\theta)$, with magnitude at least $1/4 > b/2$. Here we used $\delta, \theta \leq 1/2$ and $\eta < 1$. These cases also cover $E = 0$ and $E = I$.

Final comparison. On the bad event, one block contributes more than $(b/2)^{2p}$. After excluding this block, at most r two-dimensional blocks remain, each contributing at least $-2a^{2p}$; every remaining one-dimensional block contributes a nonnegative amount. Thus all other blocks together contribute at least $-2ra^{2p}$, so

$$\text{tr } A^{2p} + 2ra^{2p} > (b/2)^{2p} = (\theta\eta/2)^{2p}.$$

On the complementary event the indicator is zero, and the lower trace bound makes the right-hand side nonnegative. This proves (4). $\square$

The event bound in Lemma 2 applies to both spectral edges because it depends on $|\lambda - \theta|$. Its right-hand side is nonnegative after the explicit compensation, even on the event where the Gram matrix is close to I_r .

Remark (Centered projection traces). Kunisky [11, Theorem 1.12 and Sections 4–5] also passes from centered alternating traces to a spectral edge. His theorem assumes fixed positive limiting rank proportions, one equal to $1/2$, and convergence of the empirical spectral distribution to the MANOVA law. Bounds at moment orders growing faster than the logarithm of the dimension then give convergence in probability of the largest eigenvalue. The transfer uses recursions for moments of the positive compressed operator. In Lemma 2, the fixed product of the two roots instead gives a pointwise compensation for negative trace contributions and controls both edges under $\delta \leq \theta\eta^2/64$.

We now return to the Walsh projection $P = XX^T$, with $X = HD_2HD_1V$ and $\delta = r/n$. With the independent Bernoulli selector E from the start of this section, put

$$\kappa = n\theta, \quad K_2 = 4^{24}, \quad K_0 = 576K_2.$$

Proposition 3. *For an integer $p \geq 2$, suppose*

$$0 < \theta \leq \frac{1}{2}, \quad \kappa \geq r, \quad r \geq 2(4p+1)^{1000}. \quad (7)$$

Then

$$|\mathbb{E} \operatorname{tr} A^{2p}| \leq K_0^p (\delta\theta)^p. \quad (8)$$

Notice that the absolute value in (8) is outside the expectation.

The proof of Proposition 3 is given in Section 4, using the cumulant estimates in Section 3. Combining Proposition 3 with Lemma 2 gives the following Bernoulli embedding estimate.

Lemma 4. *Let*

$$p = \lceil \log_2(3000r) \rceil, \quad r \geq 2(4p+1)^{1000}.$$

For $0 < \eta < 1$, $0 < \theta \leq 1/2$, and $\kappa = n\theta \geq 64K_0r/\eta^2$, the exact random projection P and independent Bernoulli coordinate projection E satisfy

$$\mathbb{P}\{\|\theta^{-1}X^T EX - I_r\|_2 > \eta\} \leq 1/1000. \quad (9)$$

Proof. Let

$$\mathcal{B} = \{\|\theta^{-1}X^T EX - I_r\|_2 > \eta\}.$$

Since $\kappa = n\theta$ and $\delta = r/n$, the assumption $\kappa \geq 64K_0r/\eta^2$ implies

$$\kappa \geq r, \quad \delta = \frac{r\theta}{\kappa} \leq \frac{\theta\eta^2}{64K_0} \leq \frac{\theta\eta^2}{64}.$$

Hence Proposition 3 and Lemma 2 both apply. Taking expectations in the pointwise inequality (4) gives

$$\mathbb{P}(\mathcal{B}) \left(\frac{\theta\eta}{2}\right)^{2p} \leq \mathbb{E} \operatorname{tr} A^{2p} + 2ra^{2p}.$$

Using Proposition 3, $a^2 \leq \delta\theta$, and $\mathbb{E} \operatorname{tr} A^{2p} \leq |\mathbb{E} \operatorname{tr} A^{2p}|$, we obtain

$$\mathbb{P}(\mathcal{B}) \left(\frac{\theta\eta}{2}\right)^{2p} \leq (K_0^p + 2r)(\delta\theta)^p.$$

Since $\delta\theta = \frac{r\theta^2}{\kappa}$, division by $(\theta\eta/2)^{2p}$ yields

$$\mathbb{P}(\mathcal{B}) \leq (K_0^p + 2r) \left(\frac{4r}{\kappa\eta^2}\right)^p.$$

The width assumption gives $\frac{4r}{\kappa\eta^2} \leq \frac{1}{16K_0}$, and therefore, since $K_0 \geq 1$,

$$\mathbb{P}(\mathcal{B}) \leq (1 + 2r)16^{-p} \leq 3r 2^{-p}.$$

Finally, $p = \lceil \log_2(3000r) \rceil$ implies $2^{-p} \leq (3000r)^{-1}$, so

$$\mathbb{P}(\mathcal{B}) \leq 1/1000.$$

□

Lemma 4 supplies the large-rank Bernoulli estimate. The remaining sampling and rank cases will be treated in Section 5. We next prove the entry estimates needed for Proposition 3.

3 Joint cumulants of the transformed projection

To prove Proposition 3, we will expand the trace into products of entries of the centered projection $R = P - \delta I_n$. These entries are dependent. Joint cumulants organize their joint moments into contributions from blocks of entries: the second cumulant is covariance, and higher cumulants separate a collection's dependence from products of smaller subcollections. Our target is Lemma 7, which bounds joint cumulants of the scaled entries nP_{ij} by r times a factor depending only on their order and gives a Walsh-symmetry condition for their vanishing. The same estimates apply to R at orders at least two; its first cumulants vanish by centering.

We keep the frame V fixed throughout, as in Theorem 1. Section 3.1 proves the deterministic graph-sum bound that supplies the factor r . Section 3.2 develops the moment and product cumulant identities. The moment identity will expand the trace moments, while the product identity will turn the sign expansion of P into connected graph sums. Section 3.3 combines the graph bound with these identities, then uses Walsh symmetry and computes the mean to obtain the estimates needed for R .

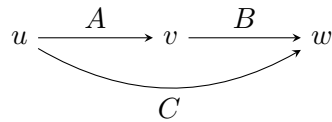
3.1 Graph contractions

The sign expansion will leave deterministic sums of matrix-entry products with shared indices. We first explain how a graph records such a sum, then prove the bound needed for the graphs that arise.

A graph contraction. For positive integers N_u, N_v, N_w , let $A \in \mathbb{R}^{N_v \times N_u}$, $B \in \mathbb{R}^{N_w \times N_v}$, and $C \in \mathbb{R}^{N_w \times N_u}$ be fixed matrices. Consider

$$S = \sum_{i=1}^{N_u} \sum_{j=1}^{N_v} \sum_{k=1}^{N_w} C_{ki} B_{kj} A_{ji}.$$

The shared indices are recorded by the following triangle:



A *labeling* $(i_u, i_v, i_w) = (i, j, k)$ selects the summand $C_{ki} B_{kj} A_{ji}$. Assigning A to $u \rightarrow v$ means that this edge contributes A_{ji} : its column index is at u and its row index is at v . The edge matrices and coordinate ranges are fixed; the labeling varies. The *graph contraction* sums the products over all

labelings. Thus the graph records shared indices among matrix-entry factors; the edge matrices are not adjacency matrices. Here contraction means index summation, not contracting an edge of the graph.

We fix the general convention for later use.

Definition 3.1 (Graph contraction). We use the real-coordinate form of a *graph of matrices* in Mingo and Speicher [14, Definition 8 and Eq. (9), pp. 2279–2280], with optional scalar vertex weights included as diagonal factors. Let $\mathcal{V}$ be the vertex set and $\mathcal{E}$ the list of edge occurrences of a finite multigraph, allowing parallel edges and loops. Each vertex u has a coordinate range $\{1, \dots, N_u\}$ from which its label i_u is chosen, and each oriented edge $e = (u, v)$ carries a fixed matrix $M_e : \mathbb{R}^{N_u} \rightarrow \mathbb{R}^{N_v}$. A *vertex weight* is an optional fixed scalar function ω_u contributing the factor $\omega_u(i_u)$. The contraction is

$$\sum_{\substack{1 \leq i_u \leq N_u \\ u \in \mathcal{V}}} \prod_{e=(u,v) \in \mathcal{E}} (M_e)_{i_v, i_u} \prod_{u \in \mathcal{V}} \omega_u(i_u). \quad (10)$$

Parallel edges give separate factors, while a loop at u gives $(M_e)_{i_u, i_u}$. A loop counts twice toward the degree, and connectedness ignores orientation. The sum is unrestricted: labels at distinct vertices may coincide when their coordinate sets coincide. It is also signed: no absolute value is taken inside the product or the sum.

Boundary matrices. Return to the triangle and fix i, k , summing only j . This gives

$$T_{ki} = \sum_{j=1}^{N_v} C_{ki} B_{kj} A_{ji} = C_{ki} (BA)_{ki}.$$

The vertices u, w are the *boundary vertices*, whose free indices index the columns and rows of T ; v is an *internal vertex*, whose index is summed out. Thus the boundary matrix is the entrywise product of C and BA , whereas the full contraction is $S = \mathbf{1}_{N_w}^T T \mathbf{1}_{N_u}$. Here $\mathbf{1}$ denotes an all-ones vector of the indicated dimension. A weight ω_v at the internal vertex inserts $\omega_v(j)$ into the sum, replacing BA by $B \operatorname{diag}(\omega_v(1), \dots, \omega_v(N_v)) A$. This also explains how vertex weights can be absorbed into diagonal edge matrices.

Definition 3.2 (Boundary matrix). For a graph as in Definition 3.1, take all vertex weights to be one and choose distinct vertices s, t . Fix $i_s = a$ and $i_t = b$ and sum all other labels to define $T : \mathbb{R}^{N_s} \rightarrow \mathbb{R}^{N_t}$ by

$$T_{ba} = \sum_{(i_u)_{u \notin \{s, t\}}} \prod_{e=(u,v) \in \mathcal{E}} (M_e)_{i_v, i_u}, \quad i_s = a, \quad i_t = b,$$

where every summed label ranges over its coordinate set. We call s the input and t the output; summing their labels recovers the full contraction as $\mathbf{1}_{N_t}^T T \mathbf{1}_{N_s}$. For an input-output graph, this is the one-input, one-output operator defined by Mingo and Speicher [14, Eq. (11), p. 2281]. Here the finite sum defines T before any input-output hypothesis is imposed; that hypothesis is needed for the operator-norm bound.

Input-output form.

Definition 3.3 (Input-output graph). In the one-input, one-output case needed here, an *input-output graph* is a directed acyclic graph with distinct vertices s, t such that s has only outgoing edges, t has only incoming edges, and every vertex lies on a directed path from s to t [14, Definition 10, p. 2280].

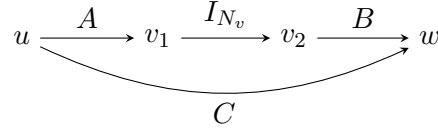
The triangle above already has this form with $s = u$ and $t = w$: its underlying undirected graph is a cycle, but its displayed orientation has no directed cycle.

For an input-output graph in Definition 3.3, Mingo and Speicher's operator bound [14, Theorem 11(1), p. 2281] states that the boundary matrix satisfies

$$\|T\|_2 \leq \prod_{e \in \mathcal{E}} \|M_e\|_2.$$

For the triangle, the right-hand side is $\|A\|_2 \|B\|_2 \|C\|_2$; there is no factor N_v from the internal sum. The original graph need not already be in input-output form. A *bridge* is an edge whose deletion disconnects a connected graph. Mingo and Speicher's graph conversion lemma [14, Lemma 16, p. 2286] states that a connected graph without bridges can be modified into input-output form with any two prescribed distinct vertices as its sole input and output. The coordinate dimensions may differ, and the edge matrices may be rectangular.

Vertex splitting. The allowed modifications are described in [14, Definition 13 and Proposition 14, p. 2285]. In the triangle example, split the internal vertex v into v_1, v_2 , attach A to v_1 and B to v_2 , and join them by the identity I_{N_v} :



The boundary entry is unchanged, because

$$\sum_{j_1, j_2=1}^{N_v} C_{ki} B_{kj_2} (I_{N_v})_{j_2 j_1} A_{j_1 i} = \sum_{j=1}^{N_v} C_{ki} B_{kj} A_{ji} = T_{ki}.$$

The identity edge enforces equality of the two new labels. This preserves each boundary entry, not only the sum S , and adds a matrix of norm one. More generally, the prescribed modifications distribute incident edges between copies of a vertex and connect the copies by identity edges. One may also reverse an edge and transpose its matrix, preserving its entry and operator norm. With the designated boundary labels held fixed, these operations preserve the boundary matrix, the full graph sum, and the product of edge norms.

Lemma 5 will use the factorization $P_0 = VV^T$ and vertex splitting to make both boundary dimensions equal to r , while the original indices range over n coordinates.

Lemma 5. *Let a connected finite multigraph have positive even degrees. Assign a matrix of operator norm at most one to each edge, with compatible coordinate dimensions at its endpoints. Suppose one edge matrix is a real rank- r orthogonal projection $P_0 = VV^T$, $r \geq 1$. At each vertex also allow a coordinate weight of absolute value at most one. The contraction of Definition 3.1, given by (10), has absolute value at most r .*

Proof. We will create two boundary vertices of dimension r , without changing the contraction. Mingo and Speicher's Lemma 16 [14] will put the graph in input-output form with these vertices as its boundaries. Their Theorem 11(1) will then bound the associated matrix $T : \mathbb{R}^r \rightarrow \mathbb{R}^r$ by $\|T\|_2 \leq 1$. The remaining factor r comes from summing the two boundary labels, as we verify at the end of the proof.

A connected even-degree graph has no bridge. Indeed, if deleting one edge separated a set of vertices from the rest, the degree sum on that set would be twice its number of internal edges plus one. This is odd, whereas a sum of even degrees is even. Represent a vertex weight $\omega_v(i_v)$ by a loop carrying $\text{diag}(\omega_v)$. Its entry at that vertex is exactly $\omega_v(i_v)$ and its norm is $\max_{i_v} |\omega_v(i_v)| \leq 1$. Thus absorbing the weights preserves every summand of the contraction. Each added loop contributes two to the degree, so the graph remains connected with positive even degrees.

The projection factorization supplies a small coordinate space. Choose the columns of V to be an orthonormal basis of range P_0 . Then $V^T V = I_r$, so $\|Vx\|_2 = \|x\|_2$ for $x \in \mathbb{R}^r$ and $\|V\|_2 = \|V^T\|_2 = 1$. Replace the distinguished edge entry by

$$(P_0)_{ij} = \sum_{a=1}^r V_{ia} V_{ja}.$$

This replaces the distinguished edge by a two-edge path through a new degree-two vertex of dimension r . Mingo and Speicher's Lemma 16 requires two distinct vertices for the input and output, not just this one. Split it into two dimension- r vertices, put one incident edge at each, and join them by I_r . In coordinates, the replacement is

$$(P_0)_{ij} = \sum_{a,b=1}^r V_{ia} (I_r)_{ab} V_{jb}.$$

If the distinguished projection acts on $\mathbb{R}^N$, the two rectangular matrices are $V : \mathbb{R}^r \rightarrow \mathbb{R}^N$ and $V^T : \mathbb{R}^N \rightarrow \mathbb{R}^r$. Call the new vertices s, t , with labels a, b , respectively. Both labels range over exactly r coordinates. Since $(I_r)_{ab}$ is zero unless $a = b$, summing the new labels recovers $(P_0)_{ij}$ for every fixed pair of old endpoint labels i, j . All other factors are unchanged, so the full signed sum is unchanged. The resulting graph is still connected with even degrees: the old endpoint degrees are unchanged, and each new vertex has degree two. If the original edge was a loop, the replacement is a three-edge cycle through the old vertex, so the same degree check applies. Thus the modified graph still has no bridge.

The graph is now connected and has no bridge, and s, t are distinct. These are precisely the hypotheses of Mingo and Speicher's Lemma 16 [14], which gives a modification in the input-output form of Definition 3.3, with s as its sole input and t as its sole output. The modifications reverse edges and transpose their matrices, or split vertices using identity edges. By their Proposition 14 [14], they preserve the full graph sum and the product of edge norms. Transposition does not change a norm, and every inserted identity has norm one. Splitting copies the coordinate space of a vertex, so both boundary dimensions remain r .

In this input-output graph, use Definition 3.2 to define $T : \mathbb{R}^r \rightarrow \mathbb{R}^r$ by taking T_{ba} to be the sum over all other labels with a, b fixed at s, t . Mingo and Speicher's Theorem 11(1) [14] applies to these compatible coordinate spaces and edge matrices. It gives $\|T\|_2 \leq \prod_e \|M_e\|_2 \leq 1$: the original edge matrices and weight loops have norm at most one, and the factors V, V^T and inserted identities have norm one. Summing a, b therefore gives the original full signed contraction $\mathbf{1}_r^T T \mathbf{1}_r$, where $\mathbf{1}_r$ is the all-ones vector. Cauchy-Schwarz and the definition of the operator norm give $|\mathbf{1}_r^T T \mathbf{1}_r| \leq \|\mathbf{1}_r\|_2^2 \|T\|_2 \leq r$. Here $\|\mathbf{1}_r\|_2^2 = r$; no ambient-dimension factor is introduced. $\square$

3.2 Cumulant identities

With the graph bound in hand, we next explain why connected graphs arise in a joint cumulant of projection entries. Each entry expands into products of signs. The Leonov-Shiryaev formula

for cumulants of products retains precisely the factor partitions that connect all the prescribed products [12, Eq. (IV.d), p. 323]. We introduce the partition notation and collect this formula, the moment-cumulant identity, and the needed vanishing rules in Lemma 6. All references to Leonov and Shiryaev below use the English translation.

Definition 3.4 (Joint cumulant). Let $q \geq 1$ and let $Y_1, \dots, Y_q$ be scalar random variables for which all joint moments appearing below are finite. Their joint cumulant is defined by the classical moment-partition formula [12, Eq. (II.c), p. 321]

$$\text{cum}(Y_1, \dots, Y_q) = \sum_{\pi \in \mathcal{P}([q])} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} \mathbb{E} \prod_{j \in B} Y_j. \quad (11)$$

Here $[q] = \{1, \dots, q\}$, and $\mathcal{P}([q])$ denotes the set of partitions of $[q]$ into nonempty disjoint blocks. For a partition π , $|\pi|$ is its number of blocks, and $B \in \pi$ denotes one such block. Each block contributes the joint moment $\mathbb{E}(\prod_{j \in B} Y_j)$; the outer product multiplies these moments over all blocks of π . The notation $\text{cum}(Y_j : j \in B)$ denotes the joint cumulant of the variables whose indices belong to B .

For example, the partitions of $[2]$ give

$$\text{cum}(Y_1, Y_2) = \mathbb{E}(Y_1 Y_2) - (\mathbb{E} Y_1)(\mathbb{E} Y_2).$$

Thus the second cumulant is covariance. Higher joint cumulants separate the dependence of a collection from products of its smaller subcollections.

We next express a joint cumulant of products as a sum of products of joint cumulants of the individual factors. The indices in a partition refer to occurrences in an expression, even when the same random variable occurs more than once. Let $\mathcal{I}$ be a finite set of occurrence positions, and let $(Z_a)_{a \in \mathcal{I}}$ be scalar random variables with the joint moments needed in the formulas below. Fix a partition $\tau = \{I_1, \dots, I_q\}$ of $\mathcal{I}$, and put $Y_f = \prod_{a \in I_f} Z_a$ for $1 \leq f \leq q$. Thus I_f records the positions of the factors in the f th product. The variables Z_a need not be distinct: different positions may carry the same random variable.

Definition 3.5 (Join of partitions). For partitions σ, τ of a finite nonempty set $\mathcal{I}$, the join of σ and τ , $\sigma \vee \tau$, is the partition generated by both identifications: two positions belong to the same join block if they can be connected by a chain of σ -blocks or τ -blocks. Write $\mathbf{1}_{\mathcal{I}} = \{\mathcal{I}\}$ for the one-block partition. For the prescribed product groups τ , the condition $\sigma \vee \tau = \mathbf{1}_{\mathcal{I}}$ expresses the *indecomposability* of the factor partition in Leonov and Shiryaev's terminology [12, Remark 5, p. 323].

In the product formula below, τ stays fixed, whereas σ ranges over partitions of the factor positions to specify the cumulants on the right-hand side.

Lemma 6 (Moment and product cumulant identities). *Let $q \geq 1$. All variables below are scalar random variables for which the joint moments in the stated formulas are finite. Cumulants and joins are as in Definitions 3.4 and 3.5.*

1. For $Y_1, \dots, Y_q$, the inverse, or moment-cumulant, identity [12, Eq. (I.c), p. 321] is

$$\mathbb{E} \prod_{j=1}^q Y_j = \sum_{\pi \in \mathcal{P}([q])} \prod_{B \in \pi} \text{cum}(Y_j : j \in B). \quad (12)$$

2. For a partition $\tau = \{I_1, \dots, I_q\}$ of the factor positions $\mathcal{I}$ and variables $(Z_a)_{a \in \mathcal{I}}$, the Leonov–Shiryaev formula [12, Eq. (IV.d), p. 323] reads

$$\text{cum} \left(\prod_{a \in I_1} Z_a, \dots, \prod_{a \in I_q} Z_a \right) = \sum_{\substack{\sigma \in \mathcal{P}(\mathcal{I}) \\ \sigma \vee \tau = \mathbf{1}_{\mathcal{I}}}} \prod_{B \in \sigma} \text{cum}(Z_a : a \in B). \quad (13)$$

3. Joint cumulants are multilinear. A joint cumulant vanishes when its arguments split into two nonempty independent groups. At order $q \geq 2$, adding deterministic constants to the arguments does not change the cumulant. An odd-order cumulant of a random variable symmetric about zero is zero.

The join condition says that the blocks of σ , together with the prescribed product groups, connect all q products. For instance, if a sign occurs twice in one product and once in another, it still occupies three positions in $\mathcal{I}$; the formula sums partitions of these three positions, not partitions of the single sign name. For a concrete example of the join condition, take $I_1 = \{1, 2\}$ and $I_2 = \{3, 4\}$. Then $\sigma = \{\{1, 3\}, \{2\}, \{4\}\}$ joins the two product groups through the block $\{1, 3\}$ and contributes the term

$$\text{cum}(Z_1, Z_3) \mathbb{E} Z_2 \mathbb{E} Z_4,$$

since a one-variable cumulant is its expectation. In contrast, $\sigma = \{\{1, 2\}, \{3, 4\}\}$ leaves the groups separate and is excluded. The join condition determines which terms occur; their cumulant factors may impose further restrictions or make them zero.

Proof. We use these formulas only as finite polynomial identities in the joint moments. First, consider the moment polynomial

$$M(t_1, \dots, t_q) = \mathbb{E} \prod_{j=1}^q (1 + t_j Y_j) = \sum_{B \subseteq [q]} \left(\mathbb{E} \prod_{j \in B} Y_j \right) \prod_{j \in B} t_j,$$

where $t_1, \dots, t_q$ are formal variables and the empty product is one. Since M has constant term one, its formal logarithm is

$$\log M = \sum_{j \geq 1} \frac{(-1)^{j-1}}{j} (M - 1)^j.$$

Every monomial of $M - 1$ has positive degree. Thus terms with $j > q$ cannot contribute to the coefficient of $t_1 \cdots t_q$. For $1 \leq j \leq q$, expanding $(M - 1)^j$ amounts to choosing an ordered list of j nonempty subsets of $[q]$, one from each factor. To obtain $t_1 \cdots t_q$, each index must occur exactly once in this list: the chosen subsets must be disjoint and their union must be $[q]$. They therefore form an ordered partition. Each unordered partition into j blocks has $j!$ such orderings, all contributing the same product of moments. If $[t_1 \cdots t_q]F$ denotes the coefficient of $t_1 \cdots t_q$ in a formal series F , this gives

$$[t_1 \cdots t_q](M - 1)^j = j! \sum_{\substack{\pi \in \mathcal{P}([q]) \\ |\pi| = j}} \prod_{B \in \pi} \mathbb{E} \prod_{i \in B} Y_i.$$

The logarithm supplies the factor $(-1)^{j-1}/j$, so each partition receives the coefficient $(-1)^{j-1}(j-1)!$. Summing over j shows that the coefficient of $t_1 \cdots t_q$ in $\log M$ is $\text{cum}(Y_1, \dots, Y_q)$ by (11). The same coefficient argument for any nonempty subset B gives $\text{cum}(Y_j : j \in B)$ as the coefficient of $\prod_{j \in B} t_j$ in $\log M$. Now compare the coefficient of $t_1 \cdots t_q$ in $M = \exp(\log M)$. Only products of monomials

on disjoint nonempty subsets can contribute. For a partition π , its $|\pi|!$ block orderings cancel the factorial denominator in the exponential series. The resulting coefficient is the right-hand side of (12), proving part 1.

To verify the coefficient in (13), apply (11) to the products $Y_1, \dots, Y_q$, and expand each resulting joint moment using (12) for the factors Z_a . For an outer block $C \subseteq [q]$, write U_C for the union $\bigcup_{f \in C} I_f$ of its factor positions. Since the I_f are disjoint, $\prod_{f \in C} Y_f = \prod_{a \in U_C} Z_a$. The two substitutions give

$$\begin{aligned} \text{cum}(Y_1, \dots, Y_q) &= \sum_{\pi \in \mathcal{P}([q])} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{C \in \pi} \mathbb{E} \prod_{a \in U_C} Z_a \\ &= \sum_{\pi \in \mathcal{P}([q])} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{C \in \pi} \left(\sum_{\sigma_C \in \mathcal{P}(U_C)} \prod_{B \in \sigma_C} \text{cum}(Z_a : a \in B) \right). \end{aligned}$$

For fixed π , the sets U_C , $C \in \pi$, partition $\mathcal{I}$. Choosing one inner partition σ_C for each C therefore gives a partition σ of $\mathcal{I}$ by taking all their blocks together. Conversely, a fixed σ arises this way exactly when every one of its blocks lies inside one U_C . In that case the inner partitions are uniquely determined, so the product of inner sums contributes it once. Keeping occurrence labels distinct, fix a partition σ and collect the terms with cumulant product $\prod_{B \in \sigma} \text{cum}(Z_a : a \in B)$. The admissible outer partitions are therefore precisely those that do not split a group of products already connected by σ and τ .

Put $b = |\sigma \vee \tau|$. Each of these b connected groups contains whole product groups I_f , since the join includes all identifications from τ . The admissible outer partitions are therefore exactly the ways to partition these b groups further. There are $S(b, j)$ ways to obtain j outer blocks, and each has coefficient $(-1)^{j-1} (j-1)!$ in (11). Hence the coefficient of the fixed cumulant product is

$$\sum_{j=1}^b S(b, j) (-1)^{j-1} (j-1)! = \mathbf{1}_{\{b=1\}},$$

where $S(b, j)$ is the number of partitions of a b -element set into j nonempty blocks. To see the cancellation explicitly, use

$$\sum_{b \geq j} S(b, j) \frac{z^b}{b!} = \frac{(e^z - 1)^j}{j!}.$$

Indeed, fix $j \geq 1$ and expand each factor as $e^z - 1 = \sum_{k \geq 1} z^k / k!$. Multiplication and collection by total degree give

$$\begin{aligned} (e^z - 1)^j &= \sum_{k_1, \dots, k_j \geq 1} \frac{z^{k_1 + \dots + k_j}}{k_1! \dots k_j!} \\ &= \sum_{b \geq j} \left(\sum_{\substack{k_1 + \dots + k_j = b \\ k_1, \dots, k_j \geq 1}} \frac{b!}{k_1! \dots k_j!} \right) \frac{z^b}{b!}. \end{aligned}$$

For fixed positive block sizes $k_1, \dots, k_j$ summing to b , the factor $b! / (k_1! \dots k_j!)$ counts ways to assign b distinct labels to j ordered blocks of those sizes: choose the labels of the first block, then the second, and so on. Summing over all positive size lists therefore counts all partitions of these labels into j ordered nonempty blocks. Every partition counted by $S(b, j)$ has exactly $j!$ block orderings,

even if some block sizes coincide, since its blocks are distinct subsets. Hence

$$\sum_{\substack{k_1+\dots+k_j=b \\ k_1,\dots,k_j\geq 1}} \frac{b!}{k_1!\dots k_j!} = j!S(b, j).$$

Thus the coefficient of $z^b/b!$ in $(e^z - 1)^j$ is $j!S(b, j)$; division by $j!$ proves the stated generating-function identity. Consequently,

$$\begin{aligned} \sum_{b\geq 1} \left[\sum_{j=1}^b S(b, j)(-1)^{j-1}(j-1)! \right] \frac{z^b}{b!} &= \sum_{j\geq 1} \frac{(-1)^{j-1}}{j} (e^z - 1)^j \\ &= \log(1 + (e^z - 1)) = z. \end{aligned}$$

This is the coefficient comparison in $\log(\exp z) = z$: the coefficient is one when $b = 1$ and zero when $b > 1$. Thus precisely the partitions with a connected join survive, each with coefficient one, proving part 2. The comparison is formal; each fixed coefficient involves only finitely many terms.

For part 3, multilinearity follows directly from (11), since each argument occurs in exactly one moment factor in every summand. Suppose the arguments $Y_1, \dots, Y_q$ split into two independent families indexed by nonempty disjoint sets A_1, A_2 with $A_1 \cup A_2 = [q]$. The usual logarithm argument for the vanishing of mixed cumulants [12, §3.B, lemma on p. 324] uses the same moment polynomial M . Independence gives

$$M = M_{A_1} M_{A_2}, \quad M_{A_i} = \mathbb{E} \prod_{j \in A_i} (1 + t_j Y_j), \quad i = 1, 2, \quad \log M = \log M_{A_1} + \log M_{A_2}.$$

Each logarithm on the right involves only the variables from its own family, so neither contains $t_1 \dots t_q$. Its coefficient, and hence the mixed joint cumulant, is zero. Only terms through total degree q are needed in these formal logarithms; no convergence assumption on a moment-generating function is used.

For $q \geq 2$, a deterministic argument forms an independent group from all the other arguments, so the corresponding cumulant is zero. Multilinearity then proves invariance under adding constants. Finally, if Z and $-Z$ have the same law, the order- q cumulant of Z equals $(-1)^q$ times itself by multilinearity. It is therefore zero for odd q . $\square$

3.3 Cumulants of the transformed projection

Recall that $K = (\mathbb{F}_2)^m$, $n = 2^m$, and $H_{ab} = n^{-1/2}(-1)^{\langle a, b \rangle}$ is the normalized Walsh matrix. The deterministic frame $V \in \mathbb{R}^{n \times r}$ satisfies $V^T V = I_r$, and $\delta = r/n$. All diagonal entries of D_1, D_2 are mutually independent uniform Rademacher signs. In this subsection, expectations and cumulants are taken over these two sign diagonals, with V and the matrix-entry indices fixed.

We now combine the two ingredients. Expanding the entries of P gives products of signs with deterministic matrix coefficients. The product formula in Lemma 6 groups these coefficients into connected graph sums, each bounded by r through Lemma 5. Counting the surviving partitions gives the order-dependent factor in the following estimate. We then use Walsh symmetry to restrict the contributing index combinations and compute the mean so that we can pass from P to R .

Lemma 7. For the projection $P = HD_2HD_1VV^TD_1HD_2H$, every integer $q \geq 1$, and arbitrary fixed pairs of indices $(i_f, j_f) \in K \times K$, $1 \leq f \leq q$, with repetitions allowed,

$$|\text{cum}(nP_{i_1j_1}, \dots, nP_{i_qj_q})| \leq (2q)^{12q}r. \quad (14)$$

The cumulant is zero unless

$$\sum_{f=1}^q (i_f + j_f) = 0 \quad \text{in } K. \quad (15)$$

Moreover, $\mathbb{E}P = \delta I_n$.

The sum in (15) is coordinatewise addition modulo two, or XOR. This is a necessary condition for a nonzero cumulant; it does not assert nonvanishing when the sum is zero.

Proof. We first prove the bound by expanding into connected graph sums, then use Walsh symmetry for the vanishing condition, and finally compute the mean. The combination of cumulant expansions and graph-sum bounds also occurs in the trace-cumulant analysis of Vázquez-Becerra [17, Section 4]; the calculation below keeps the entry indices fixed and the dependence on q and r explicit. Write $P_0 = VV^T$ and $\chi_i(a) = (-1)^{\langle i, a \rangle}$. For $\ell = 1, 2$ and $a \in K$, denote the diagonal signs by $\xi_{\ell, a} = (D_\ell)_{aa}$. The exact entry expansion is

$$nP_{ij} = \sum_{a,b,x,y \in K} \chi_i(a)\chi_j(b) \xi_{2,a}\xi_{2,b}\xi_{1,x}\xi_{1,y} H_{ax}(P_0)_{xy} H_{yb}. \quad (16)$$

The factor n cancels the two outer Hadamard normalizations. Both inner Hadamard matrices in (16) remain normalized.

For fixed summation indices in q copies of (16), let $\mathcal{I}$ be the set of the $4q$ sign occurrence positions. Set $\tau = \{I_1, \dots, I_q\}$, where I_f contains the occurrences of $\xi_{2,a_f}, \xi_{2,b_f}, \xi_{1,x_f}, \xi_{1,y_f}$ in the f th product. These are four positions even if some labels agree. By multilinearity, we may expand the joint cumulant into a sum over the q copies of the four indices, taking their deterministic coefficients outside the cumulant. Apply Lemma 6, in the form (13), and then sum the deterministic matrix and character coefficients. A nonzero partition has a partition α of its $2q$ middle-sign (D_2) occurrences and a partition β of its $2q$ input-sign (D_1) occurrences, all blocks even. Equal-coordinate occurrences within a block are identified. Indeed, the vanishing rules in Lemma 6 force each block to involve just one sign coordinate in one layer, and symmetry of that sign makes every odd-order cumulant zero. Different blocks may receive equal labels; no inequality between their labels is imposed. For example, two different blocks can both use $\xi_{2,a}$; their factors are still two cumulants in (13). Thus block membership imposes equalities within a block but does not specify the full equality pattern of all labels.

To display the resulting sum, write $\kappa_b = \text{cum}(\xi, \dots, \xi)$ with b arguments, where ξ is a single uniform Rademacher sign. For fixed α, β , let

$$S_{\alpha,\beta} = \sum_{\substack{a_f, b_f, x_f, y_f \in K \\ 1 \leq f \leq q}} \prod_{f=1}^q \chi_{i_f}(a_f) \chi_{j_f}(b_f) H_{a_f x_f}(P_0)_{x_f y_f} H_{y_f b_f}.$$

The prime imposes equality of coordinate labels within each block of α and of β , and imposes no inequalities between different blocks. Every block cumulant is then κ_b , where b is its size, independently of its coordinate label. Grouping the finite sums by their partitions therefore gives

$$\text{cum}(nP_{i_1j_1}, \dots, nP_{i_qj_q}) = \sum_{\substack{\alpha, \beta \text{ with even block sizes} \\ (\alpha \cup \beta) \vee \tau = \mathbf{1}_{\mathcal{I}}}} \left(\prod_{A \in \alpha} \kappa_{|A|} \right) \left(\prod_{B \in \beta} \kappa_{|B|} \right) S_{\alpha,\beta}.$$

Here $\alpha \cup \beta$ is the partition of all $4q$ occurrence positions obtained by taking the blocks in both layers. We now bound the deterministic sum $S_{\alpha,\beta}$ for each partition pair in this display.

Before the block identifications, the f th product of matrix entries describes the path

$$a_f \xleftarrow{H} x_f \xleftarrow{P_0} y_f \xleftarrow{H} b_f.$$

The connected-join condition in Definition 3.5 connects all q such paths. Indeed, the groups I_f connect the occurrences along each such path, and the blocks of α, β identify vertices between paths. Having one join block is exactly what makes their union connected after these identifications.

Each middle-sign position a_f, b_f is incident to one edge, whereas each input-sign position x_f, y_f is incident to two. Merging the positions in a block adds their degrees; an edge whose endpoints merge becomes a loop and still contributes twice. Thus the resulting graph has one vertex per middle-sign block, of degree equal to its even size, and one per input-sign block, of degree twice its size. It is connected with positive even degrees. Its matrices are normalized H 's and P_0 's, all of norm one, and it contains a rank- r projection edge P_0 from each original path. At each middle-sign vertex the outer characters multiply to a coordinate weight of modulus one; input-sign vertices have weight one. After the identifications, every block vertex has one freely summed label in K . Thus $S_{\alpha,\beta}$ is precisely an unrestricted signed graph contraction in Definition 3.1, and Lemma 5 gives $|S_{\alpha,\beta}| \leq r$. Absolute values have not been inserted into the matrix-entry products. The graph bound applies to the full signed sum; it does not bound the sum of the absolute values of its individual summands.

The order- b cumulant of a single Rademacher sign has absolute value at most b^{2b} : Definition 3.4, in the form (11), has at most b^b partitions, each factorial coefficient is at most $b! \leq b^b$, and each moment has absolute value at most one. For one layer, this gives

$$\prod_{A \in \alpha} |\kappa_{|A|}| \leq \prod_{A \in \alpha} |A|^{2|A|} \leq (2q)^{2 \sum_{A \in \alpha} |A|} = (2q)^{4q},$$

since its block sizes sum to $2q$. The same bound holds for β . A set of $2q$ positions has at most $(2q)^{2q}$ partitions: after numbering the positions, assigning each position the least number in its block encodes the partition by a function from that set to itself. Hence there are at most $(2q)^{4q}$ partition pairs, even before the even-size and connected-join restrictions. Applying the triangle inequality to the displayed partition sum gives

$$\begin{aligned} |\text{cum}(nP_{i_1 j_1}, \dots, nP_{i_q j_q})| &\leq \underbrace{(2q)^{4q}}_{\text{partition pairs}} \underbrace{(2q)^{4q}(2q)^{4q}}_{\text{two coefficient products}} \underbrace{r}_{\text{graph sum}} \\ &= (2q)^{12q} r, \end{aligned}$$

which proves (14).

For (15), we use a symmetry that preserves the law of P but can reverse the sign of the target cumulant whenever the XOR sum is nonzero. For $s \in K$, let $C_s = \text{diag}(\chi_s(i) : i \in K)$ and let T_s translate coordinates: $(T_s u)_a = u_{a+s}$ for $u \in \mathbb{R}^n$. In particular $T_s^2 = I_n$. The character identity $\chi_i(a+s) = \chi_i(a)\chi_i(s)$ gives the first two Walsh identities below by comparing matrix entries:

$$C_s H = H T_s, \quad T_s H = H C_s, \quad C_s H D_2 H D_1 V = H(T_s D_2 T_s) H(C_s D_1) V. \quad (17)$$

For the last identity, insert $T_s^2 = I_n$ next to D_2 and use the first two identities:

$$\begin{aligned} C_s H D_2 H D_1 V &= H T_s D_2 H D_1 V \\ &= H(T_s D_2 T_s) T_s H D_1 V \\ &= H(T_s D_2 T_s) H(C_s D_1) V. \end{aligned}$$

The diagonal entries of $T_s D_2 T_s$ are those of D_2 permuted by $a \mapsto a + s$. The entries of $C_s D_1$ are those of D_1 multiplied by fixed signs. Both operations preserve the independent uniform sign law within each diagonal, and the two new diagonals remain independent because each depends on a different original diagonal. Thus $C_s P C_s$ has the same law as P for the same fixed V . Entrywise,

$$(C_s P C_s)_{ij} = \chi_s(i) \chi_s(j) P_{ij} = \chi_s(i+j) P_{ij}.$$

Write $c = \text{cum}(nP_{i_1 j_1}, \dots, nP_{i_q j_q})$. Equality in law and multilinearity then give, for every $s \in K$,

$$c = \left(\prod_{f=1}^q \chi_s(i_f + j_f) \right) c = \chi_s \left(\sum_f (i_f + j_f) \right) c.$$

If $q_0 = \sum_f (i_f + j_f) \neq 0$, choose a nonzero coordinate of q_0 and let s be the corresponding binary basis vector. Then $\chi_s(q_0) = -1$, so the cumulant equals its negative and is zero. This proves (15).

Finally set $B = H D_1 P_0 D_1 H$, so that $P = H D_2 B D_2 H$. Condition on D_1 , which fixes B . The (a, b) entry of $D_2 B D_2$ is $B_{ab} \xi_{2,a} \xi_{2,b}$. Its conditional expectation is B_{aa} if $a = b$ and zero otherwise, by independence and centering of the signs. Thus

$$\mathbb{E}_{D_2}[D_2 B D_2 \mid D_1] = \text{diag}(B_{aa} : a \in K),$$

where the right side is the diagonal matrix with the indicated entries. To average these entries over D_1 , expand

$$B_{aa} = \sum_{x,y \in K} H_{ax} H_{ya} (P_0)_{xy} \xi_{1,x} \xi_{1,y}.$$

Only $x = y$ survives in expectation. Since $H_{ax}^2 = 1/n$ and $\sum_x (P_0)_{xx} = \text{tr } P_0 = r$, this gives

$$\mathbb{E}_{D_1} B_{aa} = \sum_x H_{ax}^2 (P_0)_{xx} = \frac{r}{n}.$$

Taking the two expectations in turn and using $H^2 = I_n$ now yields

$$\mathbb{E} P = H \text{diag}(\mathbb{E}_{D_1} B_{aa} : a \in K) H = H \left(\frac{r}{n} I_n \right) H = \delta I_n.$$

□

Recall $R = P - \delta I_n$. Every entry of R , including every diagonal entry, has mean zero. Cumulants of order at least two are unchanged by this centering, by Lemma 6, so (14)–(15) hold for R at those orders. The zero diagonal mean will remove singleton loop blocks in the trace expansion.

Remark (Trace cumulants and entry cumulants). The bounded trace-cumulant theorem of Vázquez-Becerra [17, Lemma 6 and Theorem 7] uses simultaneous conjugation invariance of the relative mixing matrices under every signed permutation, together with bounds on their entry moments. Its constants may depend on the fixed word lengths. The proof above uses the two independent sign layers for an arbitrary fixed V , retains a rank- r edge in each graph sum, and gives the explicit order dependence in (14). Its vanishing condition comes from Walsh character symmetry. The discrete Fourier sums in [17, Section 3.2] are instead evaluated through quadratic forms and Gauss sums; they do not supply the binary-rank count used in Section 4.2.

4 The trace-moment estimate

We now prove Proposition 3: under its hypotheses,

$$|\mathbb{E} \operatorname{tr}(RW)^{2p}| \leq K_0^p (\delta\theta)^p.$$

Besides estimating each term in the trace expansion, we must control how many terms contribute. Section 4.1 first averages the sampling variables and groups cyclic positions with equal row labels. It bounds the number of the resulting equality partitions and the loops in their graphs. Section 4.2 then expands the entries of R by cumulants. Lemma 9 counts the row labels allowed by their binary constraints, and Proposition 10 bounds the contribution of each equality partition using Lemma 7. Finally, Section 4.3 multiplies these counts and bounds and sums over their parameters. Keeping track of the lost free labels and cumulant blocks allows the rank hypothesis to absorb the powers of the moment order in the counts.

4.1 Equality patterns of the sampling variables

Recall that $R = P - \delta I_n$, $W = E - \theta I_n$, and $\delta = r/n$. The entries E_{ii} are independent Bernoulli(θ) variables, independent of D_1, D_2 , and the row labels lie in $K = (\mathbb{F}_2)^m$, with $|K| = n$. Write $w_i = W_{ii} = E_{ii} - \theta$. Matrix multiplication and cyclic reindexing of the diagonal factors give the original trace expansion

$$\mathbb{E} \operatorname{tr}(RW)^{2p} = \sum_{i_1, \dots, i_{2p} \in K} \mathbb{E}_{D_1, D_2} \left[\prod_{j=1}^{2p} R_{i_j i_{j+1}} \right] \mathbb{E}_E \left[\prod_{j=1}^{2p} w_{i_j} \right], \quad i_{2p+1} = i_1. \quad (18)$$

In the direct matrix product the j th diagonal factor is $w_{i_{j+1}}$; its product is the same as the one displayed. The two expectations separate because the selectors are independent of both sign diagonals.

Every word $(i_1, \dots, i_{2p})$ determines one equality partition π of the position set $[2p]$: positions belong to the same block if and only if their full row labels in K coincide. For a block $u \in \pi$, let $m_u = |u|$. Independence across distinct selector coordinates then gives

$$\mathbb{E}_E \prod_{j=1}^{2p} w_{i_j} = \prod_{u \in \pi} \mathbb{E} w^{m_u}, \quad w = \zeta - \theta, \quad \zeta \sim \text{Bernoulli}(\theta).$$

A singleton block contributes $\mathbb{E} w = 0$, so only partitions with $m_u \geq 2$ remain. Let $v = |\pi|$ be the number of blocks. Since $\sum_u m_u = 2p$, we have $1 \leq v \leq p$. Write

$$v = p - s, \quad 0 \leq s \leq p - 1.$$

Thus s counts the loss of blocks relative to a partition into p pairs. To display the regrouping explicitly, let u_j be the block containing position j , with $u_{2p+1} = u_1$. Choosing a word with equality partition π is equivalent to assigning distinct labels to its blocks, so (18) becomes

$$\mathbb{E} \operatorname{tr}(RW)^{2p} = \sum_{\substack{\pi \in \mathcal{P}([2p]) \\ |u| \geq 2 \text{ } (u \in \pi)}} \left(\prod_{u \in \pi} \mathbb{E} w^{m_u} \right) \sum_{\substack{\iota: \pi \rightarrow K \\ \iota \text{ injective}}} \mathbb{E}_{D_1, D_2} \left[\prod_{j=1}^{2p} R_{\iota(u_j), \iota(u_{j+1})} \right].$$

Here $\mathcal{P}([2p])$ denotes the set of all partitions of $[2p]$. This is the standard decomposition of a graph sum into injective sums over quotient graphs; see Male [13, Definition 2.5 and Eq. (2.2)]. We use

this finite combinatorial identity without assumptions of traffic independence. We will count the outer partitions in this subsection and estimate the inner entry sum in the next.

Construct the quotient multigraph of π by taking its blocks as vertices and, for each $j \in [2p]$, placing an edge occurrence between the blocks containing j and $j + 1$, with cyclic indexing. Edges are treated as undirected because R is symmetric, but their occurrences remain distinct. A position contributes one incoming and one outgoing half-edge, including when these form a loop. The cyclic word visits every block and connects each block to the next, so the quotient graph is connected. It has $2p$ edge occurrences and degrees $2m_u \geq 4$. Since $\sum_u \deg u = 2 \sum_u m_u = 4p$ and $v = p - s$, we obtain

$$\sum_u (\deg u - 4) = 4s. \quad (19)$$

Unlike the unrestricted contraction in Definition 3.1, the sum here is restricted to distinct full row labels for its v vertices. Let t count the vertices incident to a *nonloop* unordered edge of odd multiplicity, and let ℓ count loop occurrences. In particular, t counts vertices, not odd edges; a vertex is counted once even if several odd-multiplicity edges meet there. Since $t \leq v = p - s$, we have $s + t \leq p$. For a centered Bernoulli variable w and an integer $b \geq 2$, its two possible values give

$$\mathbb{E}w^b = \theta(1 - \theta)^b + (1 - \theta)(-\theta)^b.$$

Taking absolute values and factoring out $\theta(1 - \theta)$ yields

$$|\mathbb{E}w^b| \leq \theta(1 - \theta)((1 - \theta)^{b-1} + \theta^{b-1}) \leq \theta. \quad (20)$$

For the last inequality, use $(1 - \theta)^{b-1} \leq 1 - \theta$ and $\theta^{b-1} \leq \theta$. Thus the selector coefficient satisfies

$$\left| \prod_{u \in \pi} \mathbb{E}w^{m_u} \right| \leq \prod_{u \in \pi} \theta = \theta^v.$$

These are ordinary moments of equality blocks, not selector cumulants.

For example, take $p = 2$ and distinct labels a, b . The word (a, b, a, b) has blocks $\{1, 3\}, \{2, 4\}$ and gives four parallel edges between its two degree-four vertices. The word (a, a, b, b) has blocks $\{1, 2\}, \{3, 4\}$ and gives two connecting edges and one loop at each vertex. Both have $s = t = 0$, whereas their loop counts are respectively zero and two. These examples also explain why a label appearing twice produces degree four, not degree two.

The parameters s and t will control the graph count. The first controls the total degree above four by (19); the second locates vertices where nonloop occurrences cannot all be paired with parallel copies. Section 4.2 will relate those vertices to larger cumulant blocks or binary constraints in Lemma 9. We also need a bound on ℓ there, to count pairings of loop occurrences. We now specify the counting class independently of any row-label assignment.

Definition 4.1 (Equality-partition counting class). For $p \geq 2$, $0 \leq s \leq p - 1$ and $0 \leq t \leq p - s$, let $\mathcal{Q}_{p,s,t}$ be the set of partitions π of the fixed position set $[2p]$ with exactly $p - s$ blocks, every block of size at least two, and exactly t vertices incident to an unordered nonloop edge of odd multiplicity in the quotient graph just defined. The positions retain their numbers: partitions related by rotating or reversing the cycle are not identified unless they are the same set partition. Put $N_{p,s,t} = |\mathcal{Q}_{p,s,t}|$ and $L = 4p + 1$. For each π in this class, ℓ denotes the number of its loop occurrences.

The loop bound in (21) holds for every individual graph, whereas its bound on $N_{p,s,t}$ counts the entire class. In particular, $N_{p,s,t}$ counts partitions before any assignment of row labels. The

abbreviation $L = 4p + 1$ collects the factors depending on the moment order. The useful feature of the count is its dependence on $s + t$: its upper bound is $8p3^p$ times $L^{74(s+t)}$, so the additional powers of L depend only on $s + t$. This dependence will allow the later dimension factors to compensate for the graph count.

Lemma 8. *For the class in Definition 4.1, each $\pi \in \mathcal{Q}_{p,s,t}$ obeys the following loop bound, and the counting class obeys the size bound:*

$$\ell \leq 4t + 12s + 2, \quad N_{p,s,t} \leq 8p3^p L^{74(s+t)}. \quad (21)$$

Proof. The graph count uses the restricted local structure at vertices of degree four whose nonloop multiplicities are even. We first use this structure to bound the loops. We then compress doubled paths and count the retained graphs and their path lengths, followed by the traversals that recover the numbered cyclic positions.

The loop bound. Mark all odd-incident vertices and all vertices of degree greater than four, and call their union B . Every degree is even, so each vertex of degree greater than four contributes at least two to the total excess $4s$ in (19). There are therefore at most $2s$ such vertices. Moreover,

$$\sum_{u \in B} \deg u = 4|B| + \sum_{u \in B} (\deg u - 4) \leq 4(t + 2s) + 4s.$$

Consequently,

$$|B| \leq t + 2s, \quad D_B := \sum_{u \in B} \deg u \leq 4t + 12s. \quad (22)$$

Outside B , degree is four and all nonloop multiplicities are even. To list the possibilities, let ℓ_u be the number of loops at such a vertex u , and let a_{uz} count its edge occurrences to another vertex z . A loop contributes two to the degree. In a graph with more than one vertex, connectivity gives

$$4 = 2\ell_u + \sum_{z \neq u} a_{uz}, \quad a_{uz} \in 2\mathbb{Z}_{\geq 0}, \quad \sum_{z \neq u} a_{uz} > 0.$$

Hence ℓ_u is zero or one. If it is zero, the positive nonloop multiplicities sum to four; if it is one, they sum to two. The possibilities are exactly

type	ℓ_u	positive nonloop multiplicities
internal	0	(2, 2)
loop terminal	1	(2)
four-edge terminal	0	(4)

The internal type has two distinct neighbors. Each terminal type has one neighbor.

Suppose first that $B \neq \emptyset$. A four-edge terminal is directly adjacent to B . Otherwise its neighbor would also have degree four, with all four edges returning to the terminal, giving an isolated two-vertex component.

Starting instead at a loop terminal, follow its doubled edge. At each internal vertex there is just one other neighbor, so this doubled path continues without branching. If it ended at another loop terminal, the path would be an entire component disjoint from B ; it cannot close into a cycle through internal vertices, each of which already has its two neighbors. It must therefore reach B .

Two terminal paths cannot merge at an internal vertex and then continue to B : that would require a third neighbor. They therefore use disjoint half-edges at B , at least two per terminal. Let

T be the number of terminal vertices and ℓ_B the number of original loops at vertices in B . Each of these loops also uses two half-edges at B , distinct from the terminal attachments. Thus

$$2T + 2\ell_B \leq D_B, \quad T \leq D_B/2.$$

Every loop outside B is the single loop of a loop terminal. Consequently,

$$\ell = \ell_B + \#\{\text{loop terminals}\} \leq \ell_B + T \leq D_B.$$

If $B = \emptyset$, a four-edge terminal forces the whole graph to be the two-vertex graph with four parallel edges, by the same degree argument. Otherwise, in a graph with more than one vertex, every nonloop edge group is doubled. Replace each such group by a single edge and delete the loops. The resulting graph is connected, with degree two at internal vertices and degree one at loop terminals. Following edges can only close into a cycle or end in a path, since there is no vertex at which to branch. Restoring the doubled edges and loops gives a doubled cycle or a doubled path with one loop at each endpoint. In the one-vertex case, degree four forces exactly two loops. In all cases $\ell \leq 2$ when $B = \emptyset$. Together with (22), this proves the loop bound in (21).

Counting the retained graphs. For the count, put $a = s + t$ and first suppose $a \geq 1$. Then $B \neq \emptyset$: either $t > 0$ or the degree excess $4s$ is positive. We first count the possible retained graphs and their expansions. For each expanded graph, we then count the Euler transition systems that recover the partition of the cyclic positions. These encode closed traversals by pairing the incident half-edges at each vertex; see Fleischner, Genest, and Jackson [7, Section 1].

Retain B and all terminal vertices, and contract maximal chains of internal doubled-path vertices into distinguished doubled links. Only their lengths can vary along these chains; each removed vertex has degree four and the same two-neighbor structure. A link may return to the same retained vertex or be parallel to another edge; its paired half-edge ports must be retained. Direct doubled edges may be given length zero. An isolated cycle of removed vertices would be a component disjoint from the nonempty set B , so connectivity excludes it.

Here a half-edge is an incidence of one edge at one endpoint; a loop has two half-edges at its vertex. A port is the pair of half-edges belonging to the two copies of a doubled path at one retained endpoint. A returning path has two distinct ports at the same vertex. Its two contracted edges are loops, but each loop joins one half-edge of one port to one half-edge of the other. These loops created by contraction are distinct from the original loop occurrences counted by ℓ . The length records the number of removed internal vertices, not the number of edges. Expanding a link of length j therefore replaces its two edges by $j + 1$ doubled path segments.

The $p = 2$ words above illustrate the simplest equality graphs. The following example includes both a returning link and a link parallel to other edges. It will illustrate the two counting steps: encoding the graph by its links, then recovering the cyclic positions from a traversal. Consider the word

$$(b, x, y, b, y, x, b, z, c, b, c, z, b, c). \tag{23}$$

Its last letter is followed by its first. Here $p = 7$ and the five vertex multiplicities are $m_b = 5$, $m_c = 3$, and $m_x = m_y = m_z = 2$. Thus $s = 2$, all degrees are at least four, and the graph is connected.

There are two copies of each segment of b - x - y - b , two copies of each segment of b - z - c , and four direct b - c edges. Every nonloop multiplicity is even, so $t = 0$ and $B = \{b, c\}$. The vertices x, y, z are internal: each has two distinct neighbors joined to it by doubled edges and has no loop. Suppressing x, y gives a returning link of length two at b ; suppressing z gives a length-one link between b, c , parallel to the four direct edges. Figure 2 shows the two graphs. In the contracted graph, the

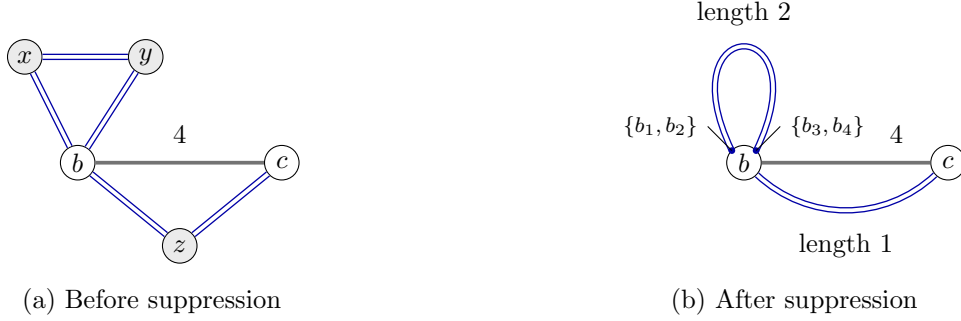


Figure 2: The graph of (23) and its contraction. Each blue double stroke represents two parallel edge occurrences; each gray segment labeled 4 represents four direct b - c edges. Suppressing x, y gives the returning link of length two, with its two ports marked at b . Suppressing z gives the parallel link of length one. Lengths count removed internal vertices.

half-edges of the two ports at b are named $\{b_1, b_2\}$ and $\{b_3, b_4\}$; the full labeling and traversal of this example are given after the proof.

We now return to the arbitrary graph with $a = s + t \geq 1$. All T terminal vertices are retained, and we have already shown that $T \leq D_B/2$. Contraction preserves the degree of every retained vertex: the two incidences of a doubled path at each endpoint are replaced by the two incidences of its link. This also holds at a returning link, whose two ports remain distinct. Each terminal has degree four. The retained graph has q vertices and total degree D , with

$$q \leq |B| + D_B/2 \leq 8a, \quad D = D_B + 4T \leq 3D_B \leq 36a, \quad q \leq p, \quad D \leq 4p. \quad (24)$$

The bounds $q \leq p$ and $D \leq 4p$ hold because only vertices were removed, while all retained degrees were preserved. Temporarily label its q vertices and distinguish their half-edges. The following data give a surjective encoding of its possible expansions: every original graph has a code. A graph may have several codes, which is harmless for an upper bound.

1. Choose the number of retained vertices and their degrees. Since $1 \leq q \leq p$, $q \leq 8a$, and every degree is at most $4p$,

$$\#\{\text{choices of } q\} \leq p \leq L^a, \quad \#\{\text{degree vectors for fixed } q\} \leq (4p + 1)^q \leq L^{8a}.$$

Retain only even degrees at least four and totals obeying the bounds on D above. The degree vector determines D and a list of D distinct half-edges, grouped by vertex.

2. Pair the half-edges to specify the edges of the retained multigraph. Pairing the first unpaired half-edge leaves $D - 1$ choices, then $D - 3$, and so on. Therefore

$$\#\{\text{edge pairings}\} = (D - 1)!! = \prod_{j=1}^{D/2} (2j - 1) \leq D^{D/2} \leq L^{18a}.$$

The last inequality uses $D \leq 4p < L$ and $D/2 \leq 18a$. Pairing two half-edges at the same vertex is allowed and produces a loop.

3. Designate disjoint pairs of half-edges at each vertex as ports. For an upper bound, give each half-edge either no mate or one of the D half-edge names as a mate. This gives

$$\#\{\text{port choices}\} \leq (D + 1)^D \leq L^{36a},$$

using $D + 1 \leq L$ and $D \leq 36a$. Retain only mutual pairings of distinct half-edges at the same vertex, and require each port to be edge-paired to one other whole port. Explicitly, after naming the half-edges of the two ports, the link has the form

$$\underbrace{\{h_1, h_2\}, \{k_1, k_2\}}_{\text{two ports}}, \quad \underbrace{\{h_1, k_1\}, \{h_2, k_2\}}_{\text{two edge pairs}}.$$

The four half-edges remain distinct even if the ports are at the same vertex. The ports identify the link even when other edges have the same endpoints.

4. Choose the length of each link and expand its internal vertices in path order. A link uses four half-edges, so there are at most $D/4 \leq 9a$ links. A length counts removed vertices and is at most p , since the original graph has $v \leq p$ vertices. Hence

$$\#\{\text{length assignments}\} \leq (p + 1)^{D/4} \leq L^{9a}.$$

Keep only expansions with the prescribed vertex and edge counts.

Multiplying the five upper bounds gives

$$\underbrace{L^a}_{\text{vertex count}} \underbrace{L^{8a}}_{\text{degrees}} \underbrace{L^{18a}}_{\text{edges}} \underbrace{L^{36a}}_{\text{ports}} \underbrace{L^{9a}}_{\text{lengths}} = L^{72a}$$

as a bound on the number of graph encodings.

For decoding, give the retained vertices the names $1, \dots, q$ and number their incident half-edges in the order of the degree vector. The edge and port pairings above then specify every retained edge and every link, including returning links.

To make the expansion deterministic, order the links by their smallest half-edge labels. On each link, start at the port with the smaller label and insert the prescribed number of internal vertices in path order. Give the new vertices and half-edges fresh names in this order. Thus the code fixes their names; no further labeling choice is needed. Undesignated half-edges keep their original edge pairings, even when they share endpoints with a designated link.

Every original graph yields such data by its specified contractions. Conversely, valid data reconstruct its graph with distinguished half-edges. Invalid and duplicate codes enlarge the count. Core labels are auxiliary; no separately labeled set of original vertices is chosen.

Recovering the cyclic positions. Now fix one expanded graph reconstructed by these data. Its edges do not yet specify the equality partition: different traversals of the same graph can place its vertices at different cyclic positions. An Euler circuit, meaning a closed traversal using every edge occurrence once, pairs arriving and departing half-edges at each vertex. These local transition pairs specify how the traversal continues through a vertex; the earlier edge pairing specified how it passes between vertices. For degree $2m_u$, pairing the first half-edge has $2m_u - 1$ choices, the next unpaired half-edge has $2m_u - 3$, and so on. Thus the number of local transition pairings is

$$(2m_u - 1)!! = 3 \prod_{j=3}^{m_u} (2j - 1) \leq 3(4p)^{m_u - 2}.$$

Here $m_u \geq 2$ and each factor $2j - 1$ is at most $4p$; when $m_u = 2$, the product on the right is empty and equals one. Multiplying over the v vertices gives

$$\prod_u (2m_u - 1)!! \leq 3^v (4p)^{\sum_u (m_u - 2)} = 3^v (4p)^{2s} \leq 3^p L^{2s},$$

where $\sum_u (m_u - 2) = 2p - 2v = 2s$ and $v \leq p$.

Choose a starting directed edge in at most $4p$ ways: there are $2p$ edge occurrences and two directions for each, including the two choices of departure half-edge for a loop. To decode, write h_j for the departing half-edge at step j and $\bar{h}_j$ for the other half-edge of that edge occurrence. The rule for continuing the traversal is

$$h_j \xrightarrow{\text{edge pairing}} \bar{h}_j \xrightarrow{\text{local pairing}} h_{j+1}.$$

Both pairings are fixed, so the initial directed edge determines every subsequent step. Local pairings can divide the edge occurrences among several circuits; retain only systems that traverse all $2p$ edge occurrences in one circuit.

For such a system, let u_j be the vertex at which h_j departs. The recovered equality partition is exactly

$$\{\{j \in [2p] : u_j = u\} : u \text{ is a vertex of the expanded graph}\}.$$

Every original equality partition produces one of these systems: its cyclic positions specify the successive edge occurrences, and each visit pairs its arriving half-edge with its departing half-edge. The edge at position 1 supplies the starting direction. Transferring these pairs to an isomorphic decoded graph therefore recovers the original partition.

Renaming the auxiliary vertices preserves every equality $u_j = u_k$, so it does not change this partition. Their names and their positions in the cyclic word are already fixed by the code and traversal; no additional vertex permutation or placement factor is required. Combining the graph and traversal counts gives

$$L^{72a} 3^p L^{2s} (4p) \leq 4p 3^p L^{74a}, \quad s \leq a.$$

For $a = 0$, the underlying graph is a doubled cycle or a doubled path with a loop at each endpoint, as in the $B = \emptyset$ case of the loop bound. Since $s = 0$, there are $v = p \geq 2$ vertices, so the one-vertex case does not arise. With p vertices fixed, there are at most these two graphs. Each vertex has exactly three local transition pairings, giving

$$N_{p,0,0} \leq \underbrace{2}_{\text{graphs}} \underbrace{4p}_{\text{starting directions}} \underbrace{3^p}_{\text{local pairings}} = 8p 3^p.$$

This proves (21). □

Decoding the example. Return to the word in (23) and its contraction in Figure 2. This example makes the encoding behind the bound on $N_{p,s,t}$ in Lemma 8 concrete. Its original graph has no loops or terminal vertices; the terminal-attachment argument for the loop bound is a separate part of the proof. The correspondence with the counting parameters is

Object	In this example	Role in the proof
Equality class	$p = 7, v = 5, s = 2, t = 0$	Fourteen positions and five letters give $s = p - v = 2$; all nonloop multiplicities are even, so the partition lies in $\mathcal{Q}_{7,2,0}$.
Marked vertices	$B = \{b, c\}$	These are the only vertices of degree greater than four; they remain when the internal vertices are removed.
Retained graph	$q = 2, D = 10 + 6 = 16$	These are the vertex and half-edge counts used in (24) and the graph encoding.
Link lengths	2 and 1	Expanding the returning link inserts x, y ; expanding the other link inserts z .

Suppressing x, y, z leaves $q = 2$ retained vertices of degrees ten and six. Label their half-edges $b_1, \dots, b_{10}$ and $c_1, \dots, c_6$. One valid retained code has the edge pairs

$$\begin{aligned} &\{b_1, b_3\}, \{b_2, b_4\}, \{b_5, c_1\}, \{b_6, c_2\}, \\ &\{b_7, c_3\}, \{b_8, c_4\}, \{b_9, c_5\}, \{b_{10}, c_6\}. \end{aligned} \quad (25)$$

Designate $\{b_1, b_2\}$ and $\{b_3, b_4\}$ as the two ports of a length-two returning link. Designate $\{b_5, b_6\}$ and $\{c_1, c_2\}$ as the ports of a length-one link between b, c . The four remaining edges have no designated ports. The first link expands to the doubled path $b-x-y-b$, and the second to $b-z-c$, parallel to the four direct $b-c$ edges. A returning link is therefore distinguished by its two ports at b , while a parallel link is distinguished from the other edges by its designated half-edges.

For actual transition data after expansion, name the fourteen edge occurrences $e_1, \dots, e_{14}$ in their traversal order in (23); repeated endpoint pairs use different copies. At each vertex the arriving/departing edge pairs are

$$\begin{aligned} b &: (e_{14}, e_1), (e_3, e_4), (e_6, e_7), (e_9, e_{10}), (e_{12}, e_{13}), \\ c &: (e_8, e_9), (e_{10}, e_{11}), (e_{13}, e_{14}), \\ x &: (e_1, e_2), (e_5, e_6), \quad y : (e_2, e_3), (e_4, e_5), \\ z &: (e_7, e_8), (e_{11}, e_{12}). \end{aligned} \quad (26)$$

The two returning-link copies in (25) expand along (e_1, e_2, e_3) and (e_6, e_5, e_4) , respectively. The copies of the other link expand along (e_7, e_8) and (e_{12}, e_{11}) . The four undesigned edges become $e_9, e_{10}, e_{13}, e_{14}$ in their listed order. Start with e_1 directed from b to x . At x , the pair (e_1, e_2) in (26) forces the next edge to be e_2 . At y , the pair (e_2, e_3) selects e_3 , which returns to b . There the pair (e_3, e_4) selects e_4 . The first steps are

$$\underbrace{b}_1 \xrightarrow{e_1} \underbrace{x}_2 \xrightarrow{e_2} \underbrace{y}_3 \xrightarrow{e_3} \underbrace{b}_4 \xrightarrow{e_4} \underbrace{y}_5.$$

The numbers beneath the vertices are positions in the cyclic word. The return to b already places positions 1, 4 in the same equality block; the two visits to y place positions 3, 5 in another. Continuing to alternate edge traversals with the specified transitions reconstructs the displayed word and hence the partition

$$\{\{1, 4, 7, 10, 13\}, \{9, 11, 14\}, \{2, 6\}, \{3, 5\}, \{8, 12\}\}.$$

All fourteen edge occurrences are traversed once. The example belongs to $\mathcal{Q}_{7,2,0}$; the auxiliary names b, c, x, y, z play no role in the resulting set partition. This is the decoding used in the counting proof: once the graph code, local transition pairs, and starting direction have been chosen, the positions belonging to each vertex require no further choice.

4.2 Cumulant partitions and binary constraints

Section 4.1 reduced the trace expansion to a sum over equality partitions and injective row-label assignments. We now fix one equality partition and bound its contribution. There are two tasks: count the cumulant partitions of its edge occurrences, and count the row labels allowed by the XOR conditions of Lemma 7. The rank of these conditions will also relate the original graph's odd-multiplicity edges to the sizes of the cumulant blocks.

Fix $\pi \in \mathcal{Q}_{p,s,t}$ as in Definition 4.1, with vertex set $\mathcal{V} = \pi$. Recall that $v = |\mathcal{V}| = p - s$, that t counts vertices incident to odd-multiplicity nonloop edges, and that ℓ counts loop occurrences. We

continue to use $L = 4p + 1$. Let u_j be the vertex containing position j , with $u_{2p+1} = u_1$. The row labels lie in $K = (\mathbb{F}_2)^m$, where $n = 2^m$; an injective assignment $\iota : \mathcal{V} \rightarrow K$ gives distinct labels to distinct vertices. Applying the moment formula (12) in Lemma 6 gives

$$\mathbb{E}_{D_1, D_2} \prod_{j=1}^{2p} R_{\iota(u_j), \iota(u_{j+1})} = \sum_{\rho \in \mathcal{P}([2p])} \prod_{C \in \rho} \text{cum}(R_{\iota(u_j), \iota(u_{j+1})} : j \in C). \quad (27)$$

The partition π groups vertex positions and fixes which row labels coincide; ρ groups edge occurrences to form cumulants. It does not identify the graph's vertices. Recall that $R = P - \delta I_n$ has centered entries, including its diagonal entries. Hence singleton blocks of ρ vanish, including singleton loops.

A partition with no singleton blocks has

$$b = p - d, \quad 0 \leq d \leq p - 1$$

blocks. Here d measures the loss of cumulant blocks from the partition into p pairs, whereas s measures the loss of vertex blocks in π .

For each cumulant block, Lemma 7 requires the sum of its endpoint labels to be zero in K . The following matrix collects these equations. Its rank counts independent restrictions on the row labels, rather than counting the equations with repetition.

Definition 4.2 (Binary constraint matrix). For an equality graph with vertex set $\mathcal{V}$ and successive vertices $u_1, \dots, u_{2p}$, with $u_{2p+1} = u_1$, and a partition ρ of its edge occurrences, the binary constraint matrix has one column per vertex and one row per block: the sum of that block's endpoint incidence vectors over $\mathbb{F}_2$. Writing $\mathbf{e}_u$ for the coordinate vector of a graph vertex, the row belonging to $C \in \rho$ is

$$\sum_{j \in C} (\mathbf{e}_{u_j} + \mathbf{e}_{u_{j+1}}) \quad \text{in } \mathbb{F}_2^{\mathcal{V}}.$$

Repeated endpoints cancel in pairs; in particular a loop contributes zero. Write h for the rank of this matrix over $\mathbb{F}_2$.

We must remove some pair blocks while the labels are still injective. Otherwise, counting all solutions of the binary system would retain partitions that actually vanish for every labeling in the trace expansion. The next lemma performs this removal and gives the two consequences of the binary constraints used below.

Lemma 9. Fix $\pi \in \mathcal{Q}_{p,s,t}$ and a partition ρ of its edge occurrences into $b = p - d$ blocks, all of size at least two. If a pair block consists of adjacent nonparallel nonloop edges, or of a loop and a nonloop edge, its cumulant vanishes for every injective row-label assignment.

After excluding partitions with either kind of pair block, the XOR conditions have exactly n^{v-h} unrestricted row-label assignments, and hence at most n^{v-h} injective ones, where $v = p - s$ and h is as in Definition 4.2. Moreover,

$$t \leq 12d + 4h. \quad (28)$$

The XOR conditions are necessary for a nonzero cumulant product; no sufficiency is asserted.

Proof. For adjacent nonparallel edges $\{u, v\}$ and $\{u, w\}$, the vertices v, w are distinct. Their binary row is

$$(\mathbf{e}_u + \mathbf{e}_v) + (\mathbf{e}_u + \mathbf{e}_w) = \mathbf{e}_v + \mathbf{e}_w.$$

Its XOR condition is $\iota(v) + \iota(w) = 0$, or $\iota(v) = \iota(w)$, which contradicts injectivity. A loop contributes zero, so pairing it with a nonloop edge $\{v, w\}$ gives the same row and the same contradiction. In both cases Lemma 7 makes the pair cumulant zero.

The remaining pair types are parallel nonloop occurrences, disjoint nonloop edges, and two loops. The first and third types have zero binary row. For disjoint edges $\{u, v\}$ and $\{w, z\}$ the four endpoints are distinct, and the row is

$$\mathbf{e}_u + \mathbf{e}_v + \mathbf{e}_w + \mathbf{e}_z.$$

Thus every nonzero pair row has weight four, meaning four nonzero coordinates.

Multiplying the row of a general block by the vertex labels gives exactly its XOR condition (15). For each of the m binary coordinates of $K = (\mathbb{F}_2)^m$, the matrix has rank h and its homogeneous system in v unknowns has 2^{v-h} solutions. The coordinates can be chosen independently, giving

$$(2^{v-h})^m = n^{v-h}.$$

Imposing injectivity can only reduce this number.

To prove (28), let M count edge occurrences in blocks of size at least three. Since there are $b = p - d$ blocks,

$$\sum_{C \in \rho} (|C| - 2) = 2p - 2b = 2d.$$

Pair blocks contribute zero to this sum. For $j \geq 3$, $j \leq 3(j - 2)$, so

$$M = \sum_{\substack{C \in \rho \\ |C| \geq 3}} |C| \leq 3 \sum_{\substack{C \in \rho \\ |C| \geq 3}} (|C| - 2) = 6d.$$

At most $2M \leq 12d$ vertices touch these occurrences.

Now consider an odd-incident vertex that touches no large-block occurrence. Choose an incident nonloop edge of odd multiplicity. All occurrences of that edge belong to pair blocks. They cannot all be paired with parallel copies, because their number is odd. At least one is paired with a different edge. The two forbidden pair types have already been excluded, so that pair is disjoint. Our vertex therefore belongs to the support of a weight-four row.

Let U be the union of the supports of these disjoint-pair rows. Choose a basis from the rows themselves. It has at most h rows, since their span is contained in the full constraint row space. Any coordinate that is zero in every basis row is zero in every linear combination. Hence all disjoint-pair rows are supported on the union of the basis supports, and

$$|U| \leq 4h.$$

Every odd-incident vertex either touches a large block or belongs to U . This gives $t \leq 12d + 4h$. $\square$

The variables introduced so far record distinct reductions in the number of free choices. Recall that $\kappa = n\theta$ is the mean sample size and $\delta = r/n$. The following table summarizes the roles of the four parameters; the exact factorization appears in (32).

Parameter	What it counts	Factor or constraint used below
$s = p - v$	Loss of selector equality blocks from the p -pair case	κ^{-s}
$d = p - b$	Loss of entry-cumulant blocks from the p -pair case	r^{-d}
h	Rank of the binary endpoint-constraint matrix	n^{-h}
t	Vertices incident to odd-multiplicity non-loop edges	$t \leq 12d + 4h$

The first two losses arise in different partitions. Identifying selector positions reduces v , while putting more edge occurrences in a cumulant reduces b . An independent binary constraint removes one freely chosen label in K . There is no separate dimensional factor for t ; instead, (28) forces every increase of t to be accounted for by d or h . This is what allows the powers of the moment order in the counts to be absorbed by the rank.

We next combine the partition count with these dimension factors. For fixed π , let $\mathcal{R}_{d,h}(\pi)$ denote the edge partitions with $p-d$ blocks of size at least two, binary rank h , and neither forbidden pair type from Lemma 9. This notation only groups the remaining terms of (27); it does not require their cumulants to be nonzero. Write

$$c(\rho) = \prod_{C \in \rho} (2|C|)^{12|C|}, \quad K_2 = 4^{24}.$$

The weight $c(\rho)$ collects the order-dependent constants in (14). Each pair block contributes K_2 . Counting with this weight lets us keep the partition count and its cumulant constants in one estimate.

Proposition 10. *Fix $\pi \in \mathcal{Q}_{p,s,t}$, and let d, h be nonnegative integers. With the notation above,*

$$\sum_{\rho \in \mathcal{R}_{d,h}(\pi)} c(\rho) \leq (3K_2)^p L^{84d+12h+9s+2t+1}. \quad (29)$$

The contribution of these partitions to the trace expansion, after summing absolute values over their injective row-label assignments and multiplying by the absolute selector coefficient, is at most

$$(3K_2)^p L^{84d+12h+9s+2t+1} (\delta\theta)^p \kappa^{-s} r^{-d} n^{-h}. \quad (30)$$

If $\mathcal{R}_{d,h}(\pi)$ is nonempty, its parameters satisfy $t \leq 12d + 4h$.

Proof. If $\mathcal{R}_{d,h}(\pi)$ is empty, both bounds are immediate. We count the large blocks first, then the disjoint pairs, loop pairs, and parallel nonloop pairs. These are all the possibilities left by Lemma 9. Throughout the count, edge occurrences retain their position numbers in $[2p]$.

Large blocks. As in the preceding proof, at most $M \leq 6d$ occurrences lie in blocks of size at least three. A subset of at most $6d$ positions can be recorded as its increasing list padded with zeros to length $6d$, giving at most $(2p+1)^{6d}$ choices. A partition of the chosen positions can be encoded by assigning each position a block label from $[2p]$, giving at most $(2p)^{6d}$ choices. Thus there are at most

$$(2p+1)^{6d} (2p)^{6d} \leq L^{12d}$$

ways to choose and partition the large-block occurrences. These codes may overcount; we retain only blocks of size at least three with the required total excess. Their cumulant constants obey

$$\prod_{\substack{C \in \rho \\ |C| \geq 3}} (2|C|)^{12|C|} \leq (4p)^{12M} \leq L^{72d}.$$

There are at most p pair blocks, whose constants together are at most K_2^p . When $d = 0$ the large-block set is empty, and both large-block counting factors equal one.

Disjoint pairs. Let U again be the union of the supports of the disjoint-pair rows. The preceding proof gives $|U| \leq 4h$. Listing these vertices in a fixed order and padding the list gives at most $(v+1)^{4h} \leq L^{4h}$ choices for U . Every occurrence in a disjoint pair has both endpoints in U . Each such occurrence uses two incidences there, so the number available is at most

$$\frac{1}{2} \sum_{u \in U} \deg u = 2|U| + \frac{1}{2} \sum_{u \in U} (\deg u - 4) \leq 2|U| + 2s \leq 8h + 2s.$$

Here all degree excesses are nonnegative, and their sum over the whole graph is $4s$ by (19). Choosing a subset of the available occurrences costs at most 2^{8h+2s} . If the chosen subset has $2j$ occurrences, pairing it costs $(2j-1)!! \leq (2p)^j \leq (2p)^{4h+s}$; the empty subset has one pairing. Therefore

$$2^{8h+2s} (2p)^{4h+s} \leq L^{8h+2s}.$$

The last inequality uses $2\sqrt{2p} \leq 4p+1 = L$. Including the choice of U gives L^{12h+2s} . We discard pairings that are not disjoint, whose support is not the chosen U , or whose resulting full binary rank is not h .

Loop pairs and parallel pairs. Once the large blocks and disjoint pairs have been chosen, every remaining loop must be paired with another loop. If their number is odd there is no such partition. Otherwise their pairing count is at most $(2p)^{\ell/2}$, since their number is at most the original loop count ℓ . The loop bound $\ell \leq 4t + 12s + 2$ from Lemma 8 gives

$$(2p)^{\ell/2} \leq L^{2t+6s+1}. \quad (31)$$

The remaining nonloop occurrences can only pair with parallel copies. For each unordered nonloop edge e , write a_e for its remaining multiplicity. An odd a_e gives no valid pairing. For positive even a_e , the number of pairings is $(a_e - 1)!!$. It equals one for $a_e = 2$ and three for $a_e = 4$; for $a_e \geq 6$,

$$(a_e - 1)!! = 3 \cdot 5 \cdot 7 \cdots (a_e - 1) \leq 3(4p)^{(a_e-4)/2}.$$

Thus the uniform bound is $3(4p)^{(a_e-4)_+/2}$, where $(x)_+ = \max\{x, 0\}$. There are at most p positive groups, because each contains at least two of the $2p$ occurrences.

We use the original degree excess to control the exponents, even though some occurrences have already been removed. At a vertex u , if no remaining group has multiplicity above four, the sum of $(a_e - 4)_+$ over incident groups is zero. Otherwise their total multiplicity is at most $\deg u$, and subtracting four for each such group subtracts at least four. In either case,

$$\sum_{e \ni u} (a_e - 4)_+ \leq \deg u - 4.$$

Every nonloop edge has two endpoints. Summing this inequality over vertices and using (19) gives

$$2 \sum_e (a_e - 4)_+ \leq \sum_u (\deg u - 4) = 4s, \quad \sum_e (a_e - 4)_+ \leq 2s.$$

All parallel-pair choices therefore cost at most

$$\prod_{e: a_e > 0} (a_e - 1)!! \leq 3^p (4p)^{\frac{1}{2} \sum_e (a_e - 4)_+} \leq 3^p L^s.$$

Combining the large-block count, cumulant constants, and the three pair counts yields

$$L^{12d} L^{72d} K_2^p L^{12h+2s} L^{2t+6s+1} 3^p L^s = (3K_2)^p L^{84d+12h+9s+2t+1},$$

which proves (29). The counts permit incompatible choices, but every partition in $\mathcal{R}_{d,h}(\pi)$ is covered. No common sign is assumed for cumulant terms.

Row labels and dimension factors. For a block of size $q \geq 2$, centering and multilinearity transfer (14) to the entries of R as

$$|\text{cum}(R_{i_1 j_1}, \dots, R_{i_q j_q})| \leq (2q)^{12q} r n^{-q}.$$

There are $b = p - d$ blocks and $2p$ occurrences in total. Their cumulant product is therefore bounded in absolute value by $c(\rho) r^{p-d} n^{-2p}$, and vanishes unless every XOR condition holds. After the forbidden pair types have been removed, Lemma 9 allows us to relax injectivity in this absolute upper bound and use at most n^{v-h} labels. The selector coefficient from (20) is at most θ^v . Their combined dimension factor is exactly

$$\theta^v r^{p-d} n^{-2p} n^{v-h} = (\delta \theta)^p \kappa^{-s} r^{-d} n^{-h}. \quad (32)$$

To see the substitutions, use $v = p - s$, $\delta = r/n$, and $\kappa = n\theta$:

$$\theta^{p-s} r^{p-d} n^{-p-s-h} = \left(\frac{r\theta}{n}\right)^p (n\theta)^{-s} r^{-d} n^{-h}.$$

Multiplying (29) by this factor proves (30). The parameter constraint is (28) in Lemma 9. $\square$

For comparison, the case $v = b = p$ and $h = 0$ contributes $\theta^p r^p n^{-p} = (\delta \theta)^p$ before its combinatorial coefficient. Reducing v by one removes one selector factor θ and one free-label factor n , hence costs κ^{-1} . Reducing b by one removes one factor r ; increasing h by one removes one factor n . This gives every exponent in (32) directly.

4.3 Summation of the trace expansion

We now combine the counts from Sections 4.1 and 4.2. The constraint $t \leq 12d + 4h$ will eliminate t , leaving three sums whose dimension factors are controlled by the rank hypothesis in Proposition 3.

Completion of the proof of Proposition 3. Recall that $A = RW$, $\delta = r/n$, and

$$L = 4p + 1, \quad \kappa = n\theta, \quad K_2 = 4^{24}, \quad K_0 = 576K_2.$$

Here $s = p - v$ and $d = p - b$ measure the losses of vertex blocks and cumulant blocks, respectively; h is the binary constraint rank, and t counts the odd-incident vertices.

For each $\pi \in \mathcal{Q}_{p,s,t}$, apply Proposition 10, whose bound (30) combines (29) with (32). Multiply this by the equality-partition count (21) from Lemma 8. The powers of L and the remaining combinatorial factors combine as follows:

$$\begin{aligned} 74(s+t) + (84d + 12h + 9s + 2t + 1) &= 83s + 76t + 84d + 12h + 1, \\ 8p 3^p (3K_2)^p &= 8p(9K_2)^p. \end{aligned}$$

Taking the last factor L outside the sum, the finite expansion and the triangle inequalities in Proposition 10 bound $|\mathbb{E} \text{tr} A^{2p}|$ by

$$8pL(9K_2)^p (\delta \theta)^p \sum_{s,t,d,h} L^{83s+76t+84d+12h} \kappa^{-s} r^{-d} n^{-h}, \quad (33)$$

where the sum runs over the feasible equality-partition parameters s, t and cumulant-partition parameters d, h . Since $v = p - s$ and $b = p - d$ lie between one and p , the block losses lie between

zero and $p - 1$. Also, $t \leq v$, and the binary matrix has b rows and v columns, so $h \leq \min\{v, b\}$. Together with Lemma 9, these observations give

$$\begin{aligned} 0 \leq s, d \leq p - 1, \quad 0 \leq t \leq p - s, \\ 0 \leq h \leq \min\{p - s, p - d\}, \quad t \leq 12d + 4h. \end{aligned}$$

For fixed s, d, h , we may enlarge the feasible t -range to all integers from zero to $T = 12d + 4h$, because every term in the upper bound is nonnegative. To sum these powers, factor out the largest one and use $L \geq 9$:

$$\sum_{t=0}^T L^{76t} = L^{76T} \sum_{j=0}^T L^{-76j} \leq \frac{L^{76T}}{1 - L^{-76}} \leq 2L^{76T}.$$

Since $76T = 912d + 304h$, this gives

$$\sum_{t=0}^{12d+4h} L^{76t} \leq 2L^{912d+304h}.$$

Thus the factor $8pL$ in (33) becomes $16pL$, and the remaining powers of L have exponents $83s + (84 + 912)d + (12 + 304)h = 83s + 996d + 316h$.

The hypotheses give $\kappa \geq r$, and $n \geq r$ because the frame has r orthonormal columns in $\mathbb{R}^n$. Since s, d, h are nonnegative and each of 83, 996, 316 is at most 1000,

$$\frac{L^{83s+996d+316h}}{\kappa^s r^d n^h} \leq \frac{L^{83s+996d+316h}}{r^{s+d+h}} \leq \left(\frac{L^{1000}}{r} \right)^{s+d+h}.$$

We can now extend all three parameter ranges to nonnegative integers, again only increasing the upper bound. Consequently (33) is at most

$$16pL(9K_2)^p(\delta\theta)^p \sum_{s,d,h \geq 0} \left(\frac{L^{1000}}{r} \right)^{s+d+h}.$$

Under (7), $r \geq 2L^{1000}$, so the common ratio is at most $1/2$. The enlarged sum factors into three convergent geometric series:

$$\sum_{s,d,h \geq 0} \left(\frac{L^{1000}}{r} \right)^{s+d+h} = \left(\sum_{j \geq 0} \left(\frac{L^{1000}}{r} \right)^j \right)^3 = \left(1 - \frac{L^{1000}}{r} \right)^{-3} \leq 8.$$

Finally

$$128pL \leq 640p^2 \leq 64^p \quad (p \geq 2);$$

the first inequality uses $L = 4p + 1 \leq 5p$. The last holds at $p = 2$, where $640p^2 = 2560 \leq 4096 = 64^p$; when p increases by one, the quantity $640p^2$ grows by a factor $((p+1)/p)^2 \leq 9/4 < 64$. Therefore

$$|\mathbb{E} \operatorname{tr} A^{2p}| \leq 128pL(9K_2)^p(\delta\theta)^p \leq (64 \cdot 9K_2)^p(\delta\theta)^p = K_0^p(\delta\theta)^p,$$

which is (8). □

5 Proof of the main theorem

Theorem 1 requires a uniform sample of a deterministic size, whereas Lemma 4 controls Bernoulli sampling and applies only above a rank threshold. We address these two restrictions separately. The first subsection transfers the Bernoulli estimate to a fixed sample size by coupling two nearby densities. The second supplies a second-moment bound for the remaining ranks. The final subsection chooses the sample size in each regime and checks the probability and width bounds of the theorem.

Recall that $X = HD_2HD_1V$ has orthonormal columns. By (1), the matrix to control is $(n/k)X^T S_J S_J^T X$, where J is a uniform k -element coordinate sample independent of the sign diagonals. Full sampling gives I_r exactly; below we estimate the error when fewer coordinates are used.

5.1 From Bernoulli sampling to a fixed sample size

The coupling below uses one ordering of independent uniform variables. Its two Bernoulli marginals give upper and lower bounds for the same fixed-size Gram matrix. For a coordinate set B , write $E_B = \text{diag}(\mathbf{1}_{\{i \in B\}} : i \in K)$; in particular, $E_J = S_J S_J^T$. We write $M \preceq N$ when $N - M$ is positive semidefinite, so this order compares every quadratic form.

Lemma 11. *Let X be an orthonormal frame, possibly random and independent of sampling. For $0 < \varepsilon < 1$, an integer $1 \leq k \leq n$, and*

$$\alpha = \varepsilon/4, \quad \theta_{\pm} = (1 \pm \alpha)k/n \leq 1,$$

suppose each normalized Bernoulli Gram of density $\theta_{\pm}$ has error at most $\varepsilon/4$ except with probability γ . Explicitly, for Bernoulli diagonal projections E_- and E_+ of the respective densities, each independent of X , the two hypotheses are

$$\begin{aligned} \mathbb{P}\{\|\theta_-^{-1}X^T E_- X - I_r\|_2 > \varepsilon/4\} &\leq \gamma, \\ \mathbb{P}\{\|\theta_+^{-1}X^T E_+ X - I_r\|_2 > \varepsilon/4\} &\leq \gamma. \end{aligned} \tag{34}$$

These are marginal probability statements; E_- and E_+ need not be independent of one another. A uniform k -element sample then satisfies

$$\mathbb{P}\left\{\left\|\frac{n}{k}X^T S_J S_J^T X - I_r\right\|_2 > \varepsilon\right\} \leq 2\gamma + 2\exp(-\varepsilon^2 k/48). \tag{35}$$

Proof. We use a common-ordering realization of the nested coupling between Bernoulli and fixed-size uniform samples described by Hájek [8, Section 2, pp. 362–363]. Two thresholds give the upper and lower samples needed for the matrix comparison. Take independent uniform U_i on $[0, 1]$, independent of X . Let J index the k smallest values, and put $B_{\pm} = \{i : U_i \leq \theta_{\pm}\}$, $N_{\pm} = |B_{\pm}|$. There are almost surely no ties. Exchangeability makes every k -element set equally likely to be J , and J is independent of X because it depends only on the U_i . For each threshold, the indicators are independent Bernoulli variables with the required mean $\theta_{\pm}$.

On $N_- \leq k \leq N_+$, every value below the lower threshold is among the first k , and all first k values are below the upper threshold. Hence

$$B_- \subseteq J \subseteq B_+, \quad E_{B_-} \preceq E_J \preceq E_{B_+}.$$

Multiplying on the left and right by X^T and X preserves the order. Since $(1 \pm \alpha)\theta_{\pm}^{-1} = n/k$, this gives

$$(1 - \alpha)\theta_-^{-1}X^T E_{B_-} X \preceq \frac{n}{k}X^T E_J X \preceq (1 + \alpha)\theta_+^{-1}X^T E_{B_+} X.$$

On the two Gram success events in (34), each normalized Bernoulli Gram lies between $(1 - \alpha)I_r$ and $(1 + \alpha)I_r$. The sandwich therefore puts the spectrum of $(n/k)X^T E_J X$ between $(1 - \alpha)^2$ and $(1 + \alpha)^2$. These endpoints are within $1 \pm \varepsilon$, because $\alpha = \varepsilon/4$ and $0 < \varepsilon < 1$ give

$$(1 - \alpha)^2 \geq 1 - 2\alpha \geq 1 - \varepsilon, \quad (1 + \alpha)^2 = 1 + \frac{\varepsilon}{2} + \frac{\varepsilon^2}{16} \leq 1 + \frac{9\varepsilon}{16} \leq 1 + \varepsilon.$$

It remains to bound the probability that the two sample counts fail to bracket k .

For a binomial N of positive mean μ and $a > 0$, we use

$$\mathbb{P}(N - \mu \geq a) \leq e^{-a^2/(2\mu+2a/3)}, \quad \mathbb{P}(N - \mu \leq -a) \leq e^{-a^2/(2\mu)}. \quad (36)$$

Here is the exponential-moment calculation. If $N \sim \text{Bin}(n, \beta)$, so $\mu = n\beta$, independence and $1 + u \leq e^u$ give

$$\mathbb{E}e^{t(N-\mu)} = e^{-t\mu}(1 - \beta + \beta e^t)^n \leq \exp(\mu(e^t - 1 - t)).$$

For $0 < t < 3$, the factorial bound $j! \geq 2 \cdot 3^{j-2}$, $j \geq 2$, gives the termwise estimate

$$e^t - 1 - t = \sum_{j \geq 2} \frac{t^j}{j!} \leq \frac{t^2}{2} \sum_{j \geq 0} (t/3)^j = \frac{t^2}{2(1 - t/3)}.$$

Markov's inequality consequently yields

$$\mathbb{P}(N - \mu \geq a) \leq \exp\left(-ta + \frac{\mu t^2}{2(1 - t/3)}\right).$$

Choose $t = a/(\mu + a/3)$, which lies in $(0, 3)$. Substitution makes the exponent $-a^2/(2\mu + 2a/3)$, proving the upper-tail bound. For the lower tail, apply the same moment bound with $-t$ and use

$$e^{-t} - 1 + t = \int_0^t (1 - e^{-u}) du \leq \int_0^t u du = \frac{t^2}{2} \quad (t \geq 0).$$

Thus

$$\mathbb{P}(N - \mu \leq -a) \leq \exp(-ta + \mu t^2/2).$$

Taking $t = a/\mu$ gives the lower-tail bound in (36).

For N_- , use $\mu = (1 - \alpha)k$ and $a = \alpha k$; for N_+ , use $\mu = (1 + \alpha)k$ and the same a . The two bounds give

$$\begin{aligned} \mathbb{P}(N_- > k) &\leq \exp\left(-\frac{\alpha^2 k}{2(1 - \alpha) + 2\alpha/3}\right) \leq e^{-\alpha^2 k/2}, \\ \mathbb{P}(N_+ < k) &\leq \exp\left(-\frac{\alpha^2 k}{2(1 + \alpha)}\right) \leq e^{-\alpha^2 k/3}, \end{aligned}$$

since $\alpha < 1/4$. The count failures therefore sum to at most $2e^{-\alpha^2 k/3}$. Union bounding these failures and the two Gram failures gives

$$\mathbb{P}\left\{\left\|\frac{n}{k}X^T E_J X - I_r\right\|_2 > \varepsilon\right\} \leq 2\gamma + 2e^{-\alpha^2 k/3} = 2\gamma + 2e^{-\varepsilon^2 k/48},$$

which is (35). This union bound uses only the two marginal Gram guarantees, as required by the statement. $\square$

5.2 A second-moment estimate for the remaining ranks

The high-order trace argument is unnecessary for bounded ranks. The next estimate averages over both sign diagonals and the uniform sample. Its quadratic dependence on r will be absorbed into the universal constant once the rank cutoff has been fixed. We estimate the Frobenius norm because its expectation can be expanded using pairs of sampled rows. The operator norm satisfies $\|M\|_2 \leq \|M\|_F$, so this also bounds the spectral error needed in Theorem 1.

Lemma 12. *For every deterministic orthonormal frame V , let $X = HD_2HD_1V$ and let $1 \leq k \leq n$ be an integer. Let J be an independent uniform k -element subset. Then*

$$\mathbb{E}_{D_1, D_2, J} \left\| \frac{n}{k} X^T S_J S_J^T X - I_r \right\|_F^2 \leq \frac{r(r+1)}{k}. \quad (37)$$

Consequently $k = \min\{n, \lceil 200r^2/\varepsilon^2 \rceil\}$ has failure probability at most 0.01.

Proof. The case $n = 1$, and every case $k = n$, is exact. For $n > 1$, let x_i^T be the rows of X , and write $G = (n/k)X^T S_J S_J^T X$. We first average over J with X fixed, then bound the fourth moments of the rows by averaging over the sign diagonals.

A uniform k -element sample has inclusion probabilities

$$\mathbb{P}(i \in J) = \frac{k}{n}, \quad \mathbb{P}(i, j \in J) = \frac{k(k-1)}{n(n-1)} \quad (i \neq j).$$

Since $G = (n/k) \sum_{i \in J} x_i x_i^T$ and $\sum_i x_i x_i^T = X^T X = I_r$, the first identity gives $\mathbb{E}[G \mid X] = I_r$. Also,

$$\sum_{i \neq j} (x_i x_i^T)(x_j x_j^T) = I_r - \sum_i \|x_i\|^2 x_i x_i^T.$$

This follows by expanding $(\sum_i x_i x_i^T)^2$ and removing the terms with $i = j$. Taking traces, the diagonal terms sum to $\sum_i \|x_i\|^4$, while the off-diagonal terms sum to $r - \sum_i \|x_i\|^4$. Applying the two inclusion probabilities to $\text{tr } G^2$ therefore yields

$$\begin{aligned} \mathbb{E}[\|G - I_r\|_F^2 \mid X] &= \mathbb{E}[\text{tr } G^2 \mid X] - r \\ &= \left(\frac{n}{k}\right)^2 \left[\frac{k}{n} \sum_i \|x_i\|^4 + \frac{k(k-1)}{n(n-1)} \left(r - \sum_i \|x_i\|^4 \right) \right] - r. \end{aligned}$$

The first equality uses symmetry of G and $\mathbb{E}[G \mid X] = I_r$. Collecting the row-moment and constant terms gives

$$\mathbb{E}[\|G - I_r\|_F^2 \mid X] = \frac{n-k}{k(n-1)} \left(n \sum_i \|x_i\|^4 - r \right). \quad (38)$$

Condition on D_1 , which fixes $U = HD_1V$, with rows u_j^T . We have $U^T U = I_r$ and $X = HD_2U$. Recalling $\xi_{2,j} = (D_2)_{jj}$, the i th row satisfies

$$\sqrt{n} x_i = \sum_j (\sqrt{n} H_{ij}) \xi_{2,j} u_j.$$

For fixed i , the factors $\sqrt{n} H_{ij}$ are deterministic signs, so multiplying by them preserves the independent Rademacher law. Thus each $\sqrt{n} x_i$ has the law of $\sum_j \xi_j u_j$, where the ξ_j are independent uniform signs.

To compute its fourth moment, expand

$$\left\| \sum_j \xi_j u_j \right\|^4 = \sum_{a,b,c,d} \xi_a \xi_b \xi_c \xi_d \langle u_a, u_b \rangle \langle u_c, u_d \rangle.$$

A sign product has zero expectation unless each index occurs an even number of times. The three pairings account for all such patterns, but count the all-equal pattern three times. Subtracting it twice gives the exact identity

$$\mathbb{E}[\xi_a \xi_b \xi_c \xi_d] = \mathbf{1}_{\{a=b\}} \mathbf{1}_{\{c=d\}} + \mathbf{1}_{\{a=c\}} \mathbf{1}_{\{b=d\}} + \mathbf{1}_{\{a=d\}} \mathbf{1}_{\{b=c\}} - 2\mathbf{1}_{\{a=b=c=d\}}.$$

The first pairing contributes $(\sum_j \|u_j\|^2)^2 = r^2$. Each of the other two contributes

$$\sum_{a,b} \langle u_a, u_b \rangle^2 = \text{tr}((U^T U)^2) = r.$$

The all-equal term contributes $\sum_j \|u_j\|^4$. Consequently

$$\mathbb{E}_\xi \left\| \sum_j \xi_j u_j \right\|^4 = r^2 + 2r - 2 \sum_j \|u_j\|^4 \leq r^2 + 2r.$$

Each row has conditional fourth moment at most $(r^2 + 2r)/n^2$. Thus the row-moment term in (38) satisfies

$$\mathbb{E}_{D_2} \left[n \sum_i \|x_i\|^4 \middle| D_1 \right] \leq r^2 + 2r.$$

Average (38) over D_2 using this bound, and then over D_1 . Since $(n-k)/(n-1) \leq 1$,

$$\mathbb{E} \|G - I_r\|_F^2 \leq \frac{n-k}{k(n-1)} (r^2 + 2r - r) \leq \frac{r(r+1)}{k},$$

which proves (37).

Finally, norm domination and Markov's inequality give

$$\mathbb{P}\{\|G - I_r\|_2 > \varepsilon\} \leq \frac{\mathbb{E} \|G - I_r\|_F^2}{\varepsilon^2} \leq \frac{r(r+1)}{k\varepsilon^2}.$$

At the stated nonfull width this is at most $(r+1)/(200r) \leq 0.01$; full width is exact. $\square$

5.3 Choice of moment order and completion

For small ranks we use the second-moment bound in Lemma 12; for large ranks we combine Lemmas 4 and 11. Full sampling covers the latter case when the ambient dimension is too small for this coupling.

We make the universal constant explicit. With K_0 as in Proposition 3, choose

$$K_0 = 576 \cdot 4^{24}, \quad R_0 = 2^{20000}, \quad C = 200R_0 + 8196K_0. \quad (39)$$

These constants establish a bound uniform in all parameters; they are not intended as practical oversampling prescriptions.

Proof of Theorem 1. Suppose first that $r < R_0$. Choose the width in Lemma 12. Its failure probability is at most 0.01, and $r^2 \leq R_0 r$ gives

$$k \leq \min\{n, \lceil 200R_0 r / \varepsilon^2 \rceil\}.$$

This is within the asserted width bound because $C \geq 200R_0$.

Now suppose $r \geq R_0$. We first verify that the integer $p = \lceil \log_2(3000r) \rceil$ satisfies the rank condition of Lemma 4. Set $x = \log_2 r \geq 20000$. Since $3000 < 2^{12}$ and taking the ceiling increases a number by less than one, $p \leq x + 13$, so $4p + 1 \leq 4x + 53$. The desired comparison follows if the function

$$f(x) = x - 1 - 1000 \log_2(4x + 53)$$

is positive. At $x = 20000$, the inequality $80053 < 2^{17}$ gives

$$f(20000) > 20000 - 1 - 1000 \cdot 17 = 2999 > 0.$$

For all $x \geq 20000$ it is increasing:

$$f'(x) = 1 - \frac{4000}{(4x + 53) \log 2} > 0,$$

because $\log 2 > 1/2$ makes the denominator greater than $80053/2 > 4000$. Exponentiating the inequality $f(x) > 0$ now gives

$$r = 2^x > 2(4x + 53)^{1000} \geq 2(4p + 1)^{1000}.$$

Thus the rank condition holds throughout this regime.

Put

$$k_0 = \left\lceil \frac{2048K_0 r}{\varepsilon^2} \right\rceil. \quad (40)$$

This width provides the Bernoulli sample sizes needed below. The comparison of n with $4k_0$ determines whether there is room for the two sampling densities in Lemma 11. If $n \leq 4k_0$, choose $k = n$. Its Gram is exact, and

$$k = n \leq 4k_0 \leq (8192K_0 + 4)r/\varepsilon^2 \leq 8196K_0 r/\varepsilon^2. \quad (41)$$

Indeed, $\lceil a \rceil \leq a + 1$ gives $4k_0 \leq 8192K_0 r/\varepsilon^2 + 4$, after which $r/\varepsilon^2 \geq 1$ and $K_0 \geq 1$ give the two displayed bounds. This case satisfies the theorem since $C \geq 8196K_0$.

It remains to consider $n > 4k_0$. Choose $k = k_0$, and take $\theta_{\pm} = (1 \pm \varepsilon/4)k_0/n$. Both densities are positive, and $k_0/n < 1/4$ gives $\theta_{\pm} \leq 5/16 < 1/2$. Their effective widths satisfy

$$n\theta_{\pm} \geq \frac{3}{4}k_0 \geq \frac{1536K_0 r}{\varepsilon^2} \geq \frac{64K_0 r}{(\varepsilon/4)^2}. \quad (42)$$

The last threshold is $1024K_0 r/\varepsilon^2$. Thus the moment order, rank condition, density range, and effective-width condition of Lemma 4 all hold with $\eta = \varepsilon/4$. Both Bernoulli failure probabilities are at most $1/1000$. Lemma 11 therefore applies with $\gamma = 1/1000$ and bounds the fixed-size failure by

$$\frac{2}{1000} + 2e^{-\varepsilon^2 k_0/48} < 0.01. \quad (43)$$

Indeed $\varepsilon^2 k_0/48 \geq 2048K_0 r/48 > 42$, and $e^{42} \geq 1 + 42 + 42^2/2 = 925 > 250$, which suffices for (43) because $2/1000 + 2/250 = 0.01$. Here $k_0 < n$ and $k_0 \leq \lceil Cr/\varepsilon^2 \rceil$, since $C \geq 2048K_0$.

All three choices of k are deterministic functions of n, r, ε , independent of V . They satisfy $k \leq n$ and the upper width bound with the single constant C in (39). They also satisfy $k \geq r$: this is immediate for $k = n$, while each nonfull choice rounds up a quantity at least r . Hence in every case

$$r \leq k \leq \min\{n, \lceil Cr/\varepsilon^2 \rceil\}.$$

For this same choice of k , the failure probability is at most 0.01 for every deterministic orthonormal frame V . Taking the supremum over these fixed frames preserves the bound. Finally, (1) identifies the Gram error with that in (2), completing the proof. $\square$

6 Concluding remarks

The quantitative estimates here prioritize a bound uniform in all parameters. It remains to determine whether sharper trace and counting estimates yield oversampling constants useful in practice. A separate structural question is how far the argument extends beyond the Walsh model. Within the present proof strategy, the ingredients to replace are the joint entry-cumulant estimates and the character constraints used here.

References

- [1] N. Amsel, Y. Baumann, P. Beckman, P. Bürgisser, C. Camaño, T. Chen, E. Chow, A. Damle, M. Dereziński, M. Embree, E. N. Epperly, R. Falgout, M. Fornace, A. Greenbaum, C. Greif, D. Halikias, Z. Huang, E. Jarlebring, Y. Koutis, D. Kressner, R. Kyng, J. Liesen, J. Lok, R. A. Meyer, Y. Nakatsukasa, K. Pearce, R. Peng, D. Persson, E. Rebrova, R. Schneider, R. Shah, E. Solomonik, N. Srivastava, A. Townsend, R. J. Webber, and J. Williams, *Linear Systems and Eigenvalue Problems: Open Questions from a Simons Workshop*, arXiv:2602.05394v3 (2026). <https://arxiv.org/abs/2602.05394v3>.
- [2] A. Andoni, P. Indyk, T. Laarhoven, I. Razenshteyn, and L. Schmidt, *Practical and optimal LSH for angular distance*, Advances in Neural Information Processing Systems **28** (2015), 1225–1233. Section numbering refers to the extended version, arXiv:1509.02897v1. <https://arxiv.org/abs/1509.02897v1>.
- [3] C. Boutsidis and A. Gittens, *Improved matrix algorithms via the subsampled randomized Hadamard transform*, SIAM Journal on Matrix Analysis and Applications **34**(3) (2013), 1301–1340. doi:10.1137/120874540. Theorem and lemma numbering refers to arXiv:1204.0062v4. <https://arxiv.org/abs/1204.0062v4>.
- [4] S. Chenakkod, M. Dereziński, and X. Dong, *Optimal oblivious subspace embeddings with near-optimal sparsity*, 52nd International Colloquium on Automata, Languages, and Programming, Leibniz International Proceedings in Informatics **334** (2025), 55:1–55:20. doi:10.4230/LIPIcs.ICALP.2025.55. Theorems 3 and 7 and Section 4 refer to the conference version. The full version is arXiv:2411.08773v2. <https://arxiv.org/abs/2411.08773v2>.
- [5] S. Chenakkod, M. Dereziński, X. Dong, and M. Rudelson, *Optimal embedding dimension for sparse subspace embeddings*, Proceedings of the 56th Annual ACM Symposium on Theory of Computing (2024), 1106–1117. doi:10.1145/3618260.3649762. Theorem numbering refers to arXiv:2311.10680v2. <https://arxiv.org/abs/2311.10680v2>.

- [6] K. Choromanski, F. Fagan, C. Gouy-Pailler, A. Morvan, T. Sarlos, and J. Atif, *TripleSpin—a generic compact paradigm for fast machine learning computations*, arXiv:1605.09046v2 (2016). <https://arxiv.org/abs/1605.09046v2>.
- [7] H. Fleischner, F. Genest, and B. Jackson, *Compatible circuit decompositions of 4-regular graphs*, Journal of Graph Theory **56**(3) (2007), 227–240. doi:10.1002/jgt.20262. Section numbering refers to the authors’ revised manuscript. <https://webpace.maths.qmul.ac.uk/b.jackson/conjRESUB.pdf>.
- [8] J. Hájek, *Limiting distributions in simple random sampling from a finite population*, Publications of the Mathematical Institute of the Hungarian Academy of Sciences **5**(3) (1960), 361–374. <https://real.mtak.hu/200979/>.
- [9] P. R. Halmos, *Two subspaces*, Transactions of the American Mathematical Society **144** (1969), 381–389. doi:10.1090/S0002-9947-1969-0251519-5.
- [10] N. Halko, P.-G. Martinsson, and J. A. Tropp, *Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions*, SIAM Review **53**(2) (2011), 217–288. doi:10.1137/090771806. Section and remark numbering refers to arXiv:0909.4061v2. <https://arxiv.org/abs/0909.4061v2>.
- [11] D. Kunisky, *Generic MANOVA limit theorems for products of projections*, arXiv:2301.09543v1 (2023). <https://arxiv.org/abs/2301.09543v1>.
- [12] V. P. Leonov and A. N. Shiryaev, *On a method of calculation of semi-invariants*, Theory of Probability and Its Applications **4**(3) (1959), 319–329 (English translation), doi:10.1137/1104031.
- [13] C. Male, *Traffic distributions and independence: Permutation invariant random matrices and the three notions of independence*, Memoirs of the American Mathematical Society **267**, no. 1300 (2020). Definition and equation numbering refers to arXiv:1111.4662v8. <https://arxiv.org/abs/1111.4662v8>.
- [14] J. A. Mingo and R. Speicher, *Sharp bounds for sums associated to graphs of matrices*, Journal of Functional Analysis **262**(5) (2012), 2272–2288. doi:10.1016/j.jfa.2011.12.010.
- [15] J. Nelson and H. L. Nguyen, *Lower bounds for oblivious subspace embeddings*, Automata, Languages, and Programming (ICALP 2014), Lecture Notes in Computer Science **8572** (2014), 883–894. doi:10.1007/978-3-662-43948-7_73. Theorem numbering refers to arXiv:1308.3280v1. <https://arxiv.org/abs/1308.3280v1>.
- [16] J. A. Tropp, *Improved analysis of the subsampled randomized Hadamard transform*, Advances in Adaptive Data Analysis **3**(1–2) (2011), 115–126. doi:10.1142/S1793536911000787. Theorem numbering refers to arXiv:1011.1595v4. <https://arxiv.org/abs/1011.1595v4>.
- [17] J. Vázquez-Becerra, *Fluctuation moments induced by conjugation with asymptotically liberating random matrix ensembles*, Journal of Theoretical Probability **36**(4) (2023), 1972–2039. doi:10.1007/s10959-023-01246-9. Lemma, theorem and section numbering refers to arXiv:2006.08902v1. <https://arxiv.org/abs/2006.08902v1>.
- [18] T. Zilca and G. Mendelson, *Approximating Uniform Random Rotations by Two-Block Structured Hadamard Rotations in High Dimensions*, arXiv:2604.23418v1 (2026). <https://arxiv.org/abs/2604.23418v1>.